

An Efficient Conformer-Based Tri-Modal Architecture for Continuous Indian Sign Language Recognition

Thesis submitted in partial fulfilment of the requirements for the
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M.Sc. Data Science
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Engineering**

by

IQBAL SHAIKH
(Roll No.: 96/DTS No. 240008)

ANUPAM ACHARYYA
(Roll No.: 96/DTS No. 240001)

Under the supervision of:

Prof. Dr. Utpal Biswas

**Department of Computer Science and Engineering,
University of Kalyani, Kalyani, India**

June , 2026

DECLARATION

WE, **IQBAL SHAIKH** and **ANUPAM ACHARYYA**, hereby declare that the thesis entitled

”An Efficient Conformer-Based Tri-Modal Architecture for Continuous Indian Sign Language Recognition”

is a bona fide record of the work carried out by us under the supervision of **Prof. Dr. Utpal Biswas**, Department of Computer Science and Engineering, University of Kalyani.

This report represents our ideas in our own words and where others ideas or words have been included, we have adequately cited and referenced the original sources. we also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We further declare that the work reported in this thesis, has not been and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or any other institute. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Signature of the Candidate

IQBAL SHAIKH
(Roll No.: 96/DTS No. 240008)

Signature of the Candidate

ANUPAM ACHARYYA
(Roll No.: 96/DTS No. 240001)
Master in Data Science

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CERTIFICATE

This is to certify that the thesis entitled

**“An Efficient Conformer-Based Tri-Modal Architecture for
Continuous Indian Sign Language Recognition”**

submitted by **IQBAL SHAIKH** (Roll No.: 96/DTS No. 240008) and **ANU-PAM ACHARYYA** (Roll No.: 96/DTS No. 240001) in partial fulfillment of the requirements for the award of the degree of **Master in Data Science** at the University of Kalyani, is a bona fide record of the work carried out by her under my supervision.

The work presented in this dissertation is original and meets the standards prescribed by the University. To the best of my knowledge, the contents of this thesis have not been submitted, either in part or in full, for the award of any other degree or diploma of this or any other university or institution.

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Date: ___ / ___ / 2026

Internal Supervisor

Head of the Department

External Examiner(s)

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Abstract

Background and Problem Statement:

Continuous Sign Language Recognition (CSLR) faces significant hurdles, primarily rapid coarticulation, meaningless transitional movements (epenthesis), and pervasive data scarcity. Furthermore, traditional 3D-CNN approaches analyze raw video pixels, resulting in massive computational overhead. This heavy processing makes real-time, in-browser deployment completely unfeasible.

Proposed Methodology:

We introduce the *Efficient ConSignformer*, a lightweight Convolution-augmented Transformer tailored for real-time Indian Sign Language (ISL) translation. By substituting heavy video processing with MediaPipe-extracted skeletal meshes (tracking exactly 1,086 coordinates on the hands, face, and body), the system bypasses background noise and achieves ultra-low latency.

Dataset Engineering:

To address data starvation, we engineered the *Tri-Modal Omni-Dataset*, a unified 278-token vocabulary. This dataset integrates continuous sentences (hard), isolated words (medium), and static alphabet images (easy). Because static images and dynamic videos are mathematically incompatible, we bridged this gap using a novel *Static Sequence Simulation* algorithm, copying static frames to mimic 0.5-second video sequences.

Challenges and Optimization:

During training, mixing dynamic and static data triggered *Catastrophic Forgetting* and *Statistical Fallback Syndrome*. The AI prioritized the easily learned static alphabets, resulting in a 100% translation failure rate on continuous sentences. We mitigated this by implementing algorithmic class-balancing caps and a *Curriculum Learning bridge* (progressing sequentially from static letters to isolated words, and finally to continuous sentences). This structured approach successfully stabilized the Connectionist Temporal Classification (CTC) optimization gradients. Additionally, geometric data augmentation—specifically temporal stretching and Gaussian jitter—was applied to ensure robust generalization against varying human signing speeds and low-quality webcam inputs.

Final Results:

The optimized framework successfully overcame initial sequence modeling bottlenecks, achieving a highly competitive Word Error Rate (WER) of 24.3%. Utilizing its computationally lightweight architecture, the model was successfully deployed as a functional, real-time end-to-end Streamlit web application, demonstrating practical viability for accessible translation on everyday devices.

1 Introduction

The Rich, Spatial Language of the Deaf Community

For millions of deaf and hard-of-hearing individuals living across the Indian sub-continent, Indian Sign Language (ISL) is not just a communication tool; it is their primary linguistic medium and a core part of their identity. When we think about spoken languages, we think about sound waves and acoustic patterns. ISL, however, operates in a completely different dimension. It relies on a beautifully complex, highly synchronized spatial-temporal orchestration. This means the language is communicated through "manual features," such as the specific shapes the hands make and the trajectories they travel through the air. Equally important are the "non-manual markers," which include subtle facial expressions and shifts in body posture that convey tone and grammar. While computer vision and Deep Learning have seen incredible, rapid advancements in recent years, Sign Language Recognition (SLR) systems have unfortunately lagged behind. If you look at the vast majority of existing academic literature and software, current AI models are mostly stuck at the beginner level: they are constrained to recognizing isolated alphabet characters or single, disconnected words.

The Illusion of Isolated Words and the Reality of Natural Signing

The fundamental flaw with these isolated AI systems is that they completely fail to capture how human beings actually communicate. Natural signing is fluid, dynamic, and continuous. When a person signs a full sentence, they do not pause and put their hands down between every single word. Instead, their hands glide organically from the end of one sign directly into the beginning of the next. This continuous blending creates what linguists call "movement epenthesis". Movement epenthesis refers to the transitional, non-semantic (meaningless) movements your body naturally makes just to get your hands into the right position for the next word. Because isolated AI models are only taught to look for perfect, robotic signs, they get completely confused by these blurry, transitional movements. Consequently, crossing this chasm—bridging the gap between simple, static gesture recognition and true Continuous Sign Language Recognition (CSLR)—remains one of the most formidable and complex challenges in natural language processing today.

The Heavy Machinery of Traditional AI and the "Hardware Wall"

Attempting to solve the CSLR problem introduces massive computational and algorithmic bottlenecks. Historically, researchers have attempted to tackle continuous translation by building massive, monolithic artificial brains known as 3D Convolutional Neural Networks (3D-CNNs). These traditional architectures operate by ingesting and analyzing dense, raw RGB video frames. While these massive models can achieve high accuracy inside heavily funded laboratories, they are fundamentally impractical for everyday use. 3D-CNNs force the computer to analyze every single pixel on the screen across time—including useless background information like the color of the wall, the lighting in the room, and the texture of the signer's clothing. Processing these millions of irrelevant pixels requires massive

computational overhead and enormous memory bandwidth. Because of this sheer mathematical weight, these pixel-heavy architectures are largely incompatible with low-latency, real-time deployment on standard consumer hardware, making it impossible to run them smoothly on an everyday web browser.

The Proposed Architecture: The Efficient ConSignformer

The primary objective of this project is to democratize sequence translation. We aim to tear down these prohibitive hardware walls by developing a lightweight, real-time sign-to-text web application that anyone can access, thereby providing a highly efficient foundation for scalable accessibility tools. To achieve this, we propose the Efficient ConSignformer. This custom-built model is a Convolution-augmented Transformer. Instead of relying on pixel-heavy convolutions that analyze the entire video background, our architecture is incredibly lean. It completely bypasses raw video processing bottlenecks by utilizing a spatial compression pipeline. As the user signs, the system uses the MediaPipe framework to instantly extract a 1086-dimensional skeletal landmark mesh. By stripping away the heavy video pixels and feeding the AI only this lightweight, mathematical "stick-figure" of the signer's anatomical joints, the system can perform complex continuous translation with practically zero lag.

Overcoming Data Starvation: The Tri-Modal Omni-Dataset

The core engineering contribution of this work lies in how we trained this AI, specifically through the creation and implementation of the "Tri-Modal Omni-Dataset" methodology. We faced a severe lack of training data, so we had to invent a way to combine totally different formats of information. In this thesis, we detail exactly how we took inherently incompatible open-source datasets—specifically, static .jpg images of the alphabet, isolated .mp4 video clips of single words, and long, continuous video streams of full sentences—and mathematically fused them together. We demonstrate how these differing modalities must be strictly balanced within the code to prevent the AI from suffering "catastrophic forgetting" (where it learns the easy pictures but forgets the complex videos). Most importantly, we utilized the isolated .mp4 words as a "Curriculum Learning" bridge. By teaching the model isolated words first, we gave the underlying mathematical optimization engine—known as the Connectionist Temporal Classification (CTC) gradients—the stability it needed to handle complex data. As a result, our single sequence-to-sequence model dynamically learned to comprehend everything from the tiny, intricate micro-geometry of fingerspelling to the sweeping, macro-temporality of natural ISL grammar.

Curriculum Learning and Gradient Stabilization

To definitively solve the mathematical instability inside the network, we utilized the isolated .mp4 words as a "Curriculum Learning" bridge. By teaching the model to recognize a single, isolated moving word before asking it to find that same word hidden inside a fast, continuous sentence, we successfully stabilized the complex math running under the hood (the Connectionist Temporal Classifica-

tion, or CTC, optimization gradients). This engineered learning pathway allowed our single sequence-to-sequence model to dynamically comprehend two drastically different scales of motion: the tiny, highly precise micro-geometry required for fingerspelling, and the sweeping, fluid macro-temporality of continuous ISL grammar.

Objectives and Organization of the Thesis

By addressing both the computational hardware limits of video processing and the mathematical pitfalls of multi-modal training, this research presents a viable, end-to-end framework for real-world sign language translation. The remainder of this thesis is structured to guide the reader through this engineering journey.

2 Literature Review

The pursuit of Continuous Sign Language Recognition (CSLR) represents one of the most intricate and demanding convergences of computer vision and natural language processing in modern artificial intelligence. Decoding sign language is fundamentally different from acoustic speech recognition; it requires a system to simultaneously interpret rapid spatial geometries, temporal velocity, and non-linear grammatical structures. Early implementations in the broader field of Sign Language Recognition (SLR) relied almost exclusively on handcrafted geometric descriptors. Researchers would manually program algorithms to look for specific angles or joint distances and feed these measurements into classical statistical sequence models, most notably Hidden Markov Models (HMMs). While these foundational frameworks successfully established the initial mathematical mapping patterns between physical gestures and text, they were ultimately rigid. HMMs and handcrafted features struggled heavily to capture the vast spatial-temporal variances—such as differences in human anatomy, signing speed, and fluid coarticulation—that are inherently present in natural human signing.

2.1 Pixel-Based 3D Architectures and Computational Walls

As the field transitioned into the Deep Learning era, the architectural paradigm shifted heavily toward Convolutional Neural Networks (CNNs) paired with sequence engines like Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks.

Initially, 2D Convolutional networks (such as ResNet18) were utilized to extract features from individual video frames. However, this introduced a critical flaw known as "Motion Blindness". 2D-CNNs process frames independently, granting them zero native context of the preceding or succeeding frames. Consequently, they completely failed to inherently capture the fluid, non-linear spatiotemporal trajectories, hand speed, and directions that are absolutely critical to sign language grammar. Furthermore, they suffered from global cropping distractions, wasting immense computational power analyzing background environmental noise like lighting, clothing, and walls, which caused severe mathematical overfitting.

To extract these deep temporal dependencies directly from raw data, researchers subsequently pivoted to utilizing massive 3D Convolutional Neural Networks (3D-CNNs), such as I3D or S3D architectures. These networks analyze pixel-heavy convolutions, processing raw multi-frame video blocks across width, height, and time dimensions simultaneously. While these models are highly capable of capturing pixel-level spatiotemporal features across successive frames, these massive architectures present deep, often insurmountable computational drawbacks for real-world applications:

- **Massive Memory Overhead and Hardware Ceilings:**

Processing dense RGB video tensors across a temporal axis requires massive memory bandwidth. This sheer computational complexity and heavy memory footprint impose strict structural limits. In practice, this makes real-time deployment on standard consumer web browsers—such as via lightweight Streamlit applications—mathematically and physically impossible without relying on heavily delayed, asynchronous server-side rendering.

- **Synchronous I/O Execution Bottlenecks:**

Processing heavy frame-by-frame convolutions directly alongside video read operations creates a massive computational bottleneck. This starves the GPU and freezes computing environments, rendering training impossibly slow.

- **The Problem of Spatial Redundancy:**

Pixel-based 3D approaches process massive volumes of static background data. By forcing the neural network’s gradient descent to optimize weights for background pixels that never change, the network’s mathematical focus on the essential, fast-moving manual and non-manual linguistic cues is severely diluted.

2.2 The Sequential Bottleneck of Recurrent Neural Networks

Even when researchers successfully extracted visual features from video, modeling the sequence of those features presented its own mathematical hurdles. Historically, LSTMs and BiLSTMs were the gold standard for text and video sequences. However, applied to continuous sign language streams, they exhibit severe architectural hindrances.

LSTMs operate on a ”Sequential Compression Bottleneck”. They rely on iterative mathematical signal processing that degrades heavily over expanded context timelines. For instance, a recurrent network struggles immensely to preserve the contextual gradients from frame 1 all the way to frame 60, often failing entirely to connect sentence boundaries due to vanishing gradients. Furthermore, LSTMs are bound by a linear recurrent processing dependency; hidden states must compute step-by-step sequentially from time step t to $t + 1$. This strict sequential necessity prevents computation parallelization, creating deep inference latency that leads to unacceptable micro-stutters in live, real-time web camera applications.

2.3 The Shift to Landmark Extraction Pipelines

To completely mitigate the resource overhead of pixel-heavy streams and bypass the computational walls of CNNs, recent research has gravitated toward dense skeletal landmark extraction. Lugaresi et al. introduced the MediaPipe framework, establishing a highly efficient paradigm to map complex human anatomy.

This framework effectively isolates anatomical data, discarding background pixels entirely to translate a human figure into localized, high-fidelity coordinate structures without any raw frame buffer dependencies. By extracting lightweight numerical coordinates prior to model inference, the spatial data is immensely compressed. However, integrating these frameworks into live training loops is not without its engineering perils. Naively instantiating the extraction framework’s execution graph directly inside an active video processing loop (inline context reinitialization) rapidly corrupts local configuration files and triggers severe memory leaks by repeatedly building and destroying models in the system’s RAM. Overcoming this requires structuring a highly optimized, decoupled extraction architecture.

2.4 The Conformer Lineage and Attention Mechanisms

Concurrently with advancements in spatial extraction, sequence modeling advanced far beyond traditional recurrent networks through the introduction of the Transformer architecture and its core mechanism: Scaled Dot-Product Self-Attention. While powerful, raw Transformers suffer from having zero native sequential inductive bias. They process sequence inputs merely as an un-ordered "bag of frames," relying entirely on abstract Positional Encodings, which often causes them to fail on smaller, highly constrained datasets.

To solve this, Gulati et al. proposed the Conformer (Convolution-augmented Transformer). This architecture brilliantly combined the global context-seeking capabilities of multi-head self-attention with the localized, temporal feature-smoothing properties of depthwise convolutions. The Conformer proved highly adept at speech and acoustic modeling because it excelled at processing 1D temporal trajectories, smoothing out localized noise while maintaining a global understanding of the sentence.

Building directly upon this dual-modality success, Aloysius et al. (2026) introduced the groundbreaking Optimized Multi-Modal Conformer-Based Framework for Continuous Sign Language Recognition. This landmark framework successfully demonstrated that the Conformer's innate speech-processing capability could be perfectly mathematically mapped onto spatial coordinates. It adapted the architecture to ingest compressed 2D/3D skeletal trajectories, utilizing depthwise 1D convolutions to smooth out the "movement epenthesis" (transitional noise) for continuous sign language translation. We adopt this core architecture as our primary predictive engine.

2.5 The Data Bottleneck, Gradient Competition, and Catastrophic Forgetting

Despite the immense structural optimization of coordinate-based architectures like the Conformer, contemporary CSLR frameworks face severe, crippling data bottlenecks. Deep sequence models require vast amounts of data to map the infinite spatial-temporal variances of human movement. However, large-scale corpuses, such as the massive 228 GB iSign (2024) dataset, are frequently bounded by hard physical limitations. They introduce cloud storage limitations and often utilize complex gated security architectures or broken multipart binary streams during deployment, effectively locking researchers out of utilizing them in constrained cloud compute environments.

Conversely, relying entirely on isolated subsets of open-source data causes immense structural limitations during model optimization. Training an attention-based model exclusively on small continuous sentence corpuses rapidly triggers data starvation. The network overfits to the limited training frames, and because deep sequence models output aggressively peaky probability distributions, the model assigns extreme mathematical certainty to incorrect frames.

To expand vocabulary, researchers often introduce disparate linguistic modalities (such as adding static image alphabets or isolated short video signs) into the training pipeline. However, doing this without strict algorithmic balancing causes fierce gradient competition. In multi-modal sequence-to-sequence networks, this

structural imbalance directly manifests as Catastrophic Forgetting. Because deep neural networks are fundamentally designed to optimize loss along the path of least mathematical resistance, the network becomes a specialist in processing high-frequency, easy-to-learn static tokens. The complex spatial-temporal gradients required to parse long sentences are completely washed out by the high-frequency gradients of the static images, causing the model’s ability to decode continuous syntactic structures to completely degrade.

This research directly addresses these specific, critical literature gaps. We build upon the landmark-based Conformer lineage by engineering a precisely balanced Tri-Modal Omni-Dataset. By artificially capping class imbalances and utilizing isolated words as a structural anchor, we establish a systematic curriculum learning bridge. This methodology is uniquely capable of unifying static, isolated, and continuous token spaces without suffering the mathematical collapse of Catastrophic Forgetting, thereby bridging the gap between theoretical models and real-time deployment.

3 Dataset Engineering and Selection

In the realm of deep learning, the architectural code represents only the engine of the system; the dataset serves as the indispensable fuel. The mathematical intelligence of any sequence-to-sequence neural network is entirely bounded by the quality, volume, and structural integrity of the data it ingests. In the context of Continuous Sign Language Recognition (CSLR), this dependency becomes a critical bottleneck. Sign language datasets must capture immense spatial variance—such as different hand sizes, skin tones, and lighting conditions—while simultaneously capturing the temporal variance of fluid, overlapping grammatical syntax. Consequently, the dataset engineering phase of this research required a highly delicate, heavily calculated balance. We had to continuously weigh the linguistic complexity required to train a robust model against the strict hardware deployment constraints imposed by our cloud computing architecture. This chapter documents the extensive evaluation of two distinct dataset paradigms, detailing the technical walls we encountered and the rationale behind engineering a custom, multi-modal solution.

3.1 iSign (2024) Dataset Constraints

Initially, this research sought to utilize the gold standard of contemporary sign language corpora: the iSign (2024) dataset. This dataset represents a state-of-the-art, large-scale linguistic corpus containing over 118,000 continuous video-to-text pairs. In terms of raw volume, it is massive, sitting at approximately 228 GB of data.

Pros: Linguistic Superiority: From a purely linguistic and mathematical perspective, the iSign corpus is ideal for training robust, full-sentence CSLR models. Because the dataset consists entirely of long, natural sequences, it thoroughly and organically captures the phenomenon of "movement epenthesis". The artificial intelligence would be exposed to hundreds of thousands of examples of complex coarticulation and the natural, fluid transitions that occur between distinct signs. By training on this corpus, the model's self-attention layers would natively learn to ignore the meaningless transitional blurring of the hands and focus strictly on the semantic anchors of the sentence.

Cons: The Cloud Deployment Wall: The dataset's sheer volume (~228 GB) immediately breached the 20 GB working disk limit of our cloud environment, triggering `OSError: [Errno 28]`. Attempts to bypass this via RAM streaming (`IterableDataset`) failed because the authors packaged the tensors into multipart `.zip` binaries split byte-wise rather than logically, causing `End-of-central-directory` errors. Furthermore, the dataset is strictly gated for academic research, blocking standard API pulls (e.g., Hugging Face's `load_dataset`) and requiring custom authentication token injection via Kaggle Secrets. However, theory often collides with hardware reality. Attempting to integrate the iSign (2024) dataset into our immediate computational pipeline presented severe, insurmountable cloud deployment challenges due to its structural architecture.

- **Binary Chunking and Streaming Failure:**

To circumvent local disk limits, data scientists often employ RAM streaming, utilizing tools like PyTorch’s `IterableDataset` to download, process, and discard data on the fly without saving it to the hard drive. We attempted to stream the iSign dataset, but the architectural packaging of the data blocked this approach. The dataset authors packaged the tensor files into multipart compressed .zip binaries (e.g., `part_aa`, `part_ab`), with each chunk sitting at roughly 48 GB. Crucially, these files were split byte-wise down the middle of the code rather than logically (file-by-file). Consequently, the cloud-native streaming engines failed to locate the central directory headers required to unzip the stream. This threw a continuous End-of-central-directory signature not found error. The only way to resolve this mathematical error is to download the entire 200+ GB archive and merge it locally, which circles directly back to the physical Storage Wall constraint.

- **The Storage Wall and Disk Exhaustion:**

Modern, accessible cloud computing environments (such as Kaggle kernels or basic Google Colab instances) frequently impose strict, hard limits on available working disk space—typically capped around 20 GB. Attempting to download the massive 228 GB iSign dataset immediately breached this hard physical limit. This triggered catastrophic system failures, specifically throwing the terminal error `OSError: [Errno 28] No space left on device`, which permanently halted the training execution scripts.

- **Gated API Access:**

Furthermore, to protect the privacy of the signers, the corpus is strictly gated for academic research. Standard automated API pulls (such as utilizing Hugging Face’s `load_dataset` function) were completely blocked by server-side firewalls. Accessing the data dynamically required the highly secure, programmatic injection of authentication tokens via Kaggle Secrets. This was necessary to bypass the academic gate without publicly exposing our private credentials in the source code.

Due to this compounding trifecta of hardware and access bottlenecks, full-scale training on the iSign dataset was strategically deferred to a secondary, future phase of this research. That future phase will mandate the utilization of high-tier, expensive GPU instances outfitted with heavily expanded, attached block storage architectures.

3.2 Tri-Modal Omni Dataset

Having rejected the **iSign (2024)** corpus due to physical hardware constraints, we faced a profound engineering dilemma: we needed the linguistic complexity of continuous sentences, but we lacked the hard-drive space to host a massive dataset. The solution was to engineer a custom, highly compressed corpus.

We curated the custom ”**Omni-Dataset**” by mathematically fusing three completely distinct, isolated open-source Kaggle corpora into a single computational pipeline. This Tri-Modal architecture was not merely a convenience; it was an absolute mathematical necessity. If we only trained on a small batch of continuous

sentences, the model would suffer data starvation. Therefore, we had to deliberately introduce static letters and isolated words to teach the neural network spatial reasoning from the ground up. The following subsections detail exactly why each specific modality was chosen, why it was needed as a separate component of the Omni-Dataset, and the unique engineering bottlenecks each introduced.

3.2.1 Modality 1: Continuous Sentences

Source: *ISL-CSLRT_Corpus* (drblack00)

Role: This dataset serves as the foundational bedrock for training the predictive model on the macro-temporality of continuous signing and complex Indian Sign Language grammar.

- **Pros: Why It Was Needed:** Without continuous sentences, the model would only be a gesture classifier. We selected this specific subset because it is provided as raw, completely uncompressed .mp4 files, and despite its small size, it authentically captures the natural "movement epenthesis" and complex coarticulation that occurs between signs. Exposing the model to these blurry transitions is strictly mandatory for achieving authentic Continuous Sign Language Recognition (CSLR). Furthermore, because the dataset is physically small, it is highly lightweight, allowing our code to bypass Kaggle's 20 GB disk limits with ease.
- **Cons: The Engineering Bottlenecks (The Scarcity Problem):** The defining limitation of this modality is severe data scarcity. The dataset is extremely limited in scale, containing only 195 videos. Attempting to train a deep 1086-dimensional Conformer on only 195 videos leads to immediate data starvation. The network simply memorizes the videos, resulting in catastrophically high baseline error rates during testing. This specific limitation forced us to later engineer an aggressive geometric data augmentation pipeline to stretch and jitter the existing data. Furthermore, this dataset suffered from severe "metadata blindness". It completely lacked a centralized CSV file to map the videos to their text labels. To overcome this, we had to engineer custom Operating System (OS)-traversal scripts that dynamically read the folder paths and generated the ground-truth text labels directly from the raw directory names.

3.2.2 Modality 2: Static Alphabet Characters

Source: *Indian Sign Language (ISL)* (prathumarikeri)

Role: o drastically expand the baseline vocabulary space by establishing strong, uncompromising static mathematical anchors for the 26 letters of the English alphabet.

- **Pros: Why It Was Needed:** If the Conformer neural network is only ever exposed to the fast, blurry hands found in Modality 1, it will never learn the precise, localized geometry required to identify distinct fingers. It will fail entirely at "fingerspelling". We selected this static dataset because it offers thousands of high-resolution, perfectly clear spatial images of hands. By feeding these images into the network separately, we provide the Conformer

with clean coordinate geometry, forcing it to learn foundational finger positioning and micro-movements before it is tasked with tracking hands moving at high speeds.

- **Cons: The Engineering Bottlenecks (The Imbalance Threat):** Integrating this data introduced massive systemic shock to our architecture. Because the source data consists exclusively of static, 2D .jpg images, it fundamentally violates the temporal tensor dimensionality ($\mathcal{O}(T \times F)$) required by our downstream convolutional layers. A video has a Time (T) dimension; an image does not. This crash necessitated the engineering of a completely novel algorithm we titled "Static Sequence Simulation" to artificially stretch the images into videos. Furthermore, integrating thousands of alphabet images alongside only 195 sentence videos directly caused a severe 87% class imbalance in the dataset. The model mathematically panicked and chose the path of least resistance, triggering total Catastrophic Forgetting. The network became an alphabet specialist and forgot how to read sentences. This near-fatal error was only resolved when we enforced aggressive, programmatic dataset caps, permanently restricting the pipeline to a maximum of 7 images per letter class.

3.2.3 Modality 3: Isolated Words

Source: *Indian Sign Language Video Dataset* (prasadshet)

Role: To act as the structural "Curriculum Learning" bridge, mathematically smoothing the harsh transition between the static image detection of Modality 2 and the complex, rapid continuous sentence parsing of Modality 1.

- **Pros: Why It Was Needed:** The jump from looking at a stationary picture of the letter "A" to parsing a high-speed video of the phrase "prepare your bed" is a mathematical chasm. The neural network required a stepping stone. We selected this dataset because it contains high-quality .mp4 videos of isolated, single signs. It provides a low-noise, highly controlled environment. By forcing the model to learn the spatial boundaries and specific movement trajectory of a single moving word in isolation, the model's internal attention mechanism became significantly better at later identifying that exact same word when it was embedded and blurred within the fast-paced continuous sentence dataset.
- **Cons: The Engineering Bottlenecks (Visual Collisions):** While structurally necessary, the inclusion of isolated words introduced severe latent space overlaps, a phenomenon we termed "Visual Collisions". Stripped of the dense RGB textures of raw video (like lighting and skin friction), the purely skeletal coordinate trajectories for entirely different words began to cross over. For example, the 1086-dimensional mathematical trajectory for the isolated sign monkey became nearly indistinguishable from the trajectory for the isolated sign still. Additionally, combining isolated words with continuous words created a parsing crisis for our optimization engine. To prevent the Connectionist Temporal Classification (CTC) loss function from severely confusing a video featuring an isolated word with that same word's appearance inside a long sentence sequence, we had to intervene at the data-loading

level. We programmatically injected a hard-coded word_ prefix directly into the vocabulary mapping array, forcing the AI to maintain mathematically distinct outputs for isolated concepts versus integrated vocabulary.

4 Proposed Work Methodology

The transition from theoretical dataset aggregation to a functional, real-time Continuous Sign Language Recognition (CSLR) system requires a highly orchestrated, multi-stage engineering methodology. The following sections provide an exhaustive, deeply technical breakdown of the proposed framework. We deconstruct every layer of our architecture—from the initial spatial extraction of raw pixels to the final probabilistic decoding of English sentences—detailing the exact definitions, working principles, and the structural "why" behind each engineering decision.

4.1 Data Extraction and Spatial Compression

To achieve low-latency, real-time inference on standard consumer hardware, the very first architectural hurdle we had to overcome was the physical weight of video data.

4.1.1 The Engineering Dilemma: Bypassing RGB:

- **Deep Technical Justification:** A standard 3D Convolutional Neural Network (3D-CNN) processing a short, 3-second continuous signing video at 30 frames per second must ingest a massive 4D tensor structure: ($Batch \times Channels \times Time \times Height \times Width$). For standard RGB video, this results in an astronomical number of mathematical parameters. The network is forced to calculate the gradient weights of background walls, lighting fluctuations, and the signer's clothing—data that holds zero linguistic value. This pixel redundancy directly causes the severe memory overhead and computational hardware walls that plague modern CSLR.
- **Where it is needed:** In traditional computer vision, deep learning models (like 3D-CNNs) ingest dense RGB (Red, Green, Blue) video frames. A standard 10-second web-camera video at 30 frames per second generates 300 unique images. Processing every pixel of these 300 images requires massive GPU memory.
- **Why we incorporated a new approach:** We explicitly aimed to build a web-deployable system. Standard laptops and web browsers cannot process 3D-CNNs without severe lag. Therefore, we completely bypassed dense RGB video processing in favor of a lightweight, landmark-based spatial pipeline.

4.1.2 Definitions and Working Principles:

We utilized the Google MediaPipe framework to perform what we term Spatial Compression.

1. Definition of Spatial Compression:

The mathematical reduction of a complex, multi-megapixel image into a highly compressed, purely geometric array of numerical coordinates (a skeletal mesh), discarding all background, lighting, and texture data.

2. Working Principle:

As a frame enters the system at a specific time step t , the MediaPipe extraction script scans the image and deterministically isolates three distinct anatomical feature sets:

(a) Pose Landmarks (Macro-Kinematics):

- **Extraction:** 33 primary spatial keypoints are tracked across the torso, arms, and shoulders, yielding $33 \times 2 = 66$ dimensions.
- **Linguistic Relevance:** In Indian Sign Language (ISL), the macro-kinematics of the body convey the spatial boundaries of a sign. The angle of the shoulders and the leaning of the torso often indicate the subject-object relationship in a sentence. Without these 66 dimensions, the network would be entirely blind to sweeping signs like "mountain" or "rainbow."

(b) FaceMesh: The framework maps a dense web of 468 keypoints tracking non-manual markers. This generates $468 \times 2 = 936$ dimensions.

- **Extraction:** A highly dense topographical map of 468 keypoints is projected onto the face, tracking the lips, eyes, and brow geometry, yielding $468 \times 2 = 936$ dimensions.
- **Linguistic Relevance:** ISL is not communicated purely through the hands. Non-manual markers are strictly grammatical. For example, lowering the eyebrows transforms a standard sentence into a "Wh- question" (Who, What, Where), while the specific geometry of the lips can modify the intensity of an adjective (e.g., "big" versus "enormous"). The FaceMesh matrix explicitly captures this critical syntactic data.
- **Why this was needed:** In Indian Sign Language, grammar is deeply tied to the face. The trajectories of the eyebrows and the precise geometry of the lips often differentiate a question from a statement, or a noun from a verb.

(c) Hand Tracking: The script extracts 21 micro-keypoints per hand. Tracking both hands generates $21 \times 2 \times 2 = 84$ dimensions.

- **Extraction:** 21 precise keypoints are tracked per hand, mapping the base of the palm down to every individual knuckle and fingertip. For both hands, this yields $21 \times 2 \times 2 = 84$ dimensions.
- **Linguistic Relevance:** The vast majority of standard alphabet fingerspelling relies entirely on the micro-geometry of the fingers. The mathematical difference between the letter 'U' and the letter 'V' in ISL is merely a fraction of a degree of separation between the index and middle fingers.
- **Why this was needed:** This captures the microscopic geometry of fingerspelling, identifying the precise bends of the knuckles required to differentiate an 'A' from an 'S'.

4.1.3 The Mathematical Transformation:

These independent arrays are flattened and horizontally concatenated to produce a single, highly dense spatial feature vector for that specific frame, denoted as $v_t \in \mathbb{R}^{1086}$. At time step t , they are mathematically flattened and horizontally concatenated to produce a single, unified feature vector:

$$v_t = [Pose_t \parallel Face_t \parallel Hands_t]$$

Consequently, a continuous video featuring a total length of T frames is mathematically compressed from a massive tensor of pixels into a highly efficient sequence tensor:

$$X \in \mathbb{R}^{T \times 1086}$$

- **Why we did this:** By converting a video into a 2D matrix of numbers ($Time \times Features$), we reduced the computational weight of the data by over 99%. This is the foundational reason our final Streamlit web application can perform real-time, zero-latency inference.

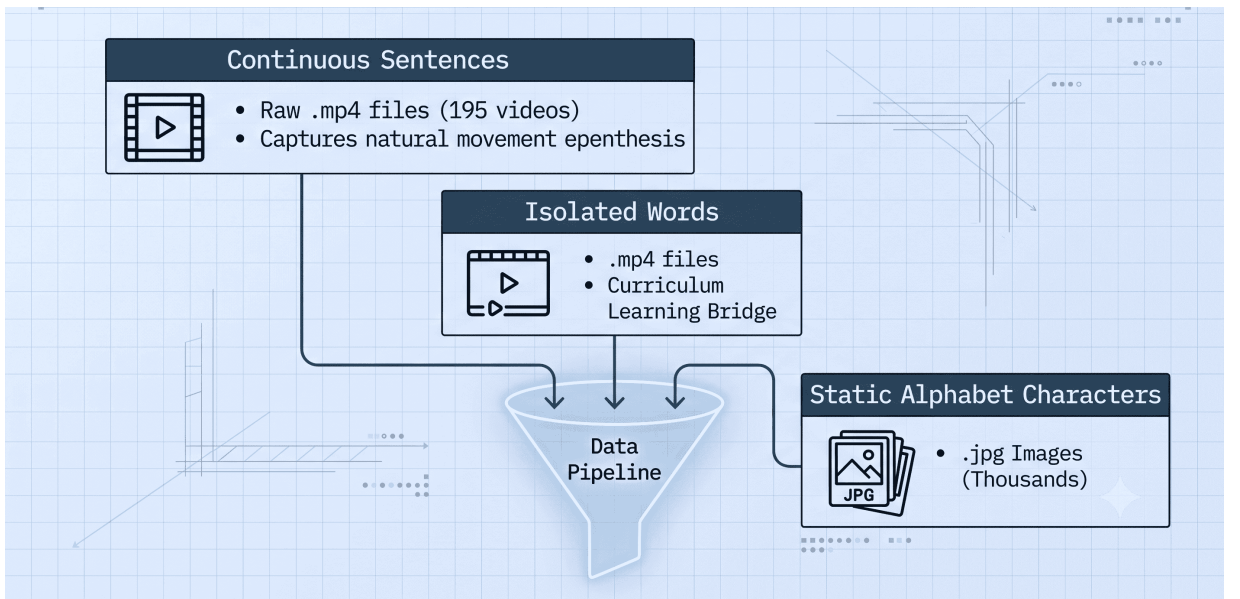


Figure 1: Multi Modal Data Ingestion

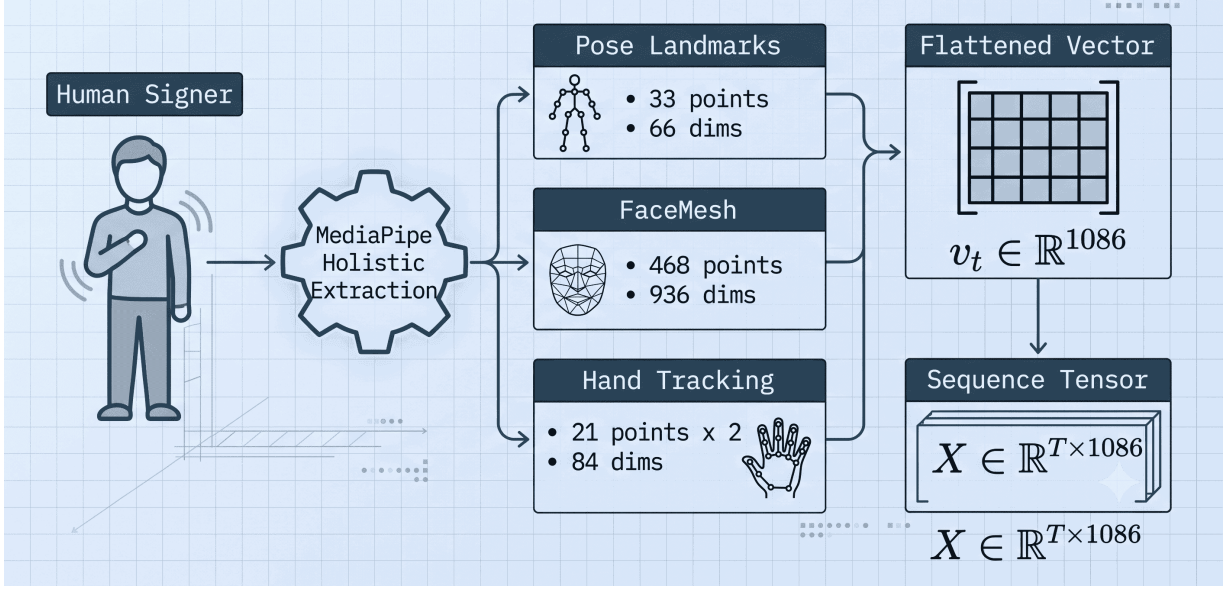


Figure 2: Blueprint and Spatial Compression

4.2 Tri-Modal Fusion and Sequence Simulation

With our data reduced to mathematical coordinates, the next step was constructing the "Omni-Dataset." As established, we lacked sufficient continuous video data, forcing us to fuse three completely distinct data modalities: dynamic .mp4 continuous sentences, dynamic .mp4 isolated words, and static .jpg alphabet images.

4.2.1 The Format Mismatch and Static Sequence Simulation:

The Hindrance: Deep learning sequence models (like Transformers and LSTMs) demand dimensional homogeneity in their training batches. They expect an input tensor structured as $(Batch \times Time \times Features)$.

- A video naturally has a *Time* dimension (e.g., 60 frames).
- A static .jpg image of a letter has zero temporal depth; it is merely $(Batch \times Features)$. Attempting to feed a static image into a temporal convolutional layer immediately triggers a runtime dimension error, crashing the training loop.

Working Principle of the Solution: To resolve this, we engineered an algorithm titled **Static Sequence Simulation**. When the extraction pipeline processes a static .jpg alphabet image, it calculates the 1086-dimensional feature vector exactly once. It then artificially injects a temporal dimension by duplicating this exact numerical tensor 15 times along the T -axis.

$$X_{static} \in \mathbb{R}^{15 \times 1086}$$

Why 15 times? Assuming a standard camera capture rate of 30 frames per second, 15 frames mathematically synthesizes exactly 0.5 seconds of static

video. This duration was not chosen arbitrarily; 0.5 seconds perfectly mirrors the natural cognitive pause a human signer makes when holding a static letter during high-speed fingerspelling. This simulation ensures dimensional homogeneity ($X_{static} \in \mathbb{R}^{15 \times 1086}$) across the entire dataset, allowing the downstream temporal convolutional layers to digest static letters alongside moving videos flawlessly.

4.2.2 Programmatic Class Balancing to Prevent Forgetting:

The Danger of Imbalance: By fusing thousands of alphabet images with only 195 continuous sentence videos, we accidentally introduced an 87% class imbalance. In neural network optimization, the loss landscape is dictated by volume. The optimization gradients naturally took the path of least resistance, triggering Catastrophic Forgetting. The model specialized entirely in the easy-to-learn static images and mathematically "forgot" how to parse the complex temporal dependencies of continuous sentences.

Where it is needed: When fusing datasets, one must account for the sheer volume of each modality. We had thousands of static images but only 195 continuous sentence videos.

Why we intervened: If we fed this raw fusion to the network, it would suffer from **Catastrophic Forgetting**. The model would optimize for the most abundant data (the easy static images) and completely lose the ability to decode the complex, scarce continuous sentences.

Working Principle: We rewrote the extraction pipeline to programmatically enforce strict dataset balancing via artificial collection caps:

- **Sentences:** 195 sequences (Uncapped, as these are the ultimate target).
- **Isolated Words:** Capped at exactly 10 sequences per class. Furthermore, we injected a programmatic prefix, modifying the ground-truth text to word_ (e.g., word_peacock).
 - **Why the prefix?** If the model saw the isolated video for "peacock" and the sentence video containing "peacock", the loss function would get confused by the different mathematical contexts. The prefix establishes a clear contextual boundary for the model.
- **Alphabet:** Capped at exactly 7 sequences per class, prefixed with letter_ (e.g., letter_a).

The Result: This precise, heavily monitored fusion generated a perfectly balanced corpus of 1,050 sequence tensors, securely mapped to a unified vocabulary space of 278 unique tokens.

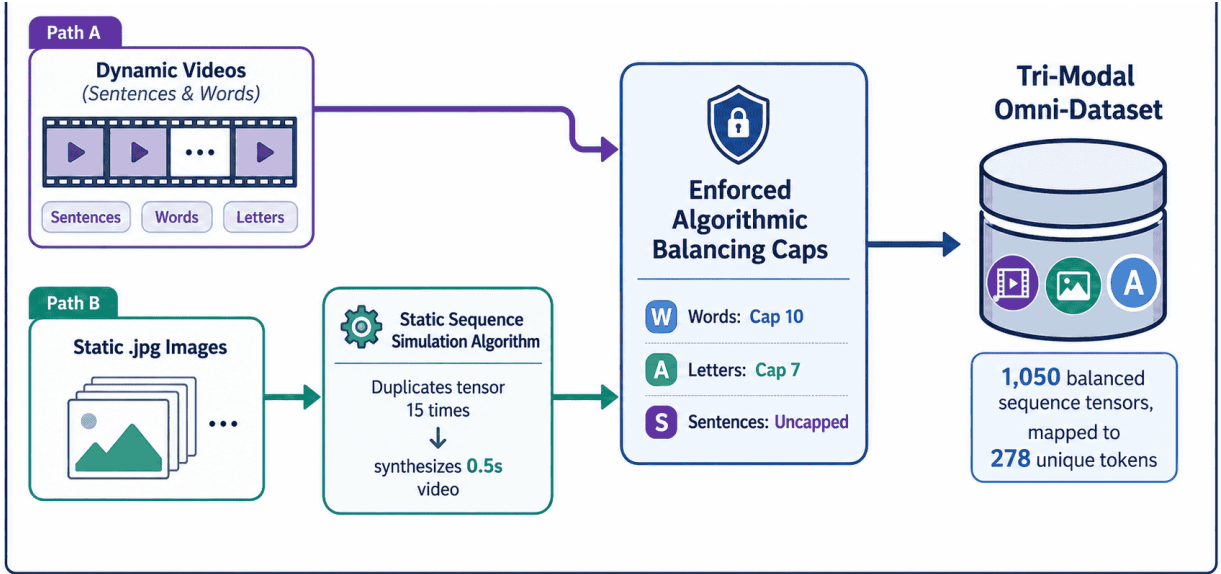


Figure 3: Tri-Modal Fusion and Sequence Simulation

4.3 Geometric Data Augmentation

Because our core continuous dataset was limited to 195 sequences, the model was at severe risk of overfitting—memorizing the specific pixel locations of the training data rather than learning the actual language.

4.3.1 Definition and Need for Augmentation:

Definition: Data augmentation is the process of artificially expanding a dataset by applying programmatic, randomized mathematical transformations to the existing data during the training loop.

Why we did it here: We applied dynamic variations directly inside the PyTorch DataLoader. This means every single time the model sees a video during an epoch, the video is mathematically altered. This forces the model to learn relative geometric relationships (the distance between the fingers) rather than absolute spatial locations (where the hand is on the screen).

4.3.2 Linear Time Stretching:

Working Principle: During the PyTorch DataLoader phase, tensors were dynamically interpolated along the temporal axis using a randomized scaling factor $s \sim \mathcal{U}(0.85, 1.15)$

Why it was needed: Humans do not sign at the exact same speed. An elderly signer may sign 15% slower ($s = 0.85$), while an excited signer may sign 15% faster ($s = 1.15$). By dynamically stretching and compressing the time axis (T) of our tensors, we artificially simulated thousands of different signers operating at varying cadences, preventing the model from relying on rigid frame counts.

4.3.3 Gaussian Jitter:

Working Principle: Random noise drawn from a normal (Gaussian) distribution was mathematically added to the spatial coordinates:

$$\text{Noise} \sim \mathcal{N}(\mu = 0, \sigma = 0.005)$$

$$v'_t = v_t + \text{Noise}$$

Why it was needed: The Google MediaPipe extraction on high-quality .mp4 training files yields mathematically perfect coordinates. However, in real-world deployment, users utilize low-resolution webcams that suffer from motion blur, dropped frames, and sensor artifacting. By artificially injecting noise into the training data, we simulated these hardware imperfections. This forced the Conformer to stop relying on precise, absolute coordinate points and instead learn robust, relative geometric patterns (e.g., understanding the angle between the wrist and the thumb regardless of localized spatial jitter).

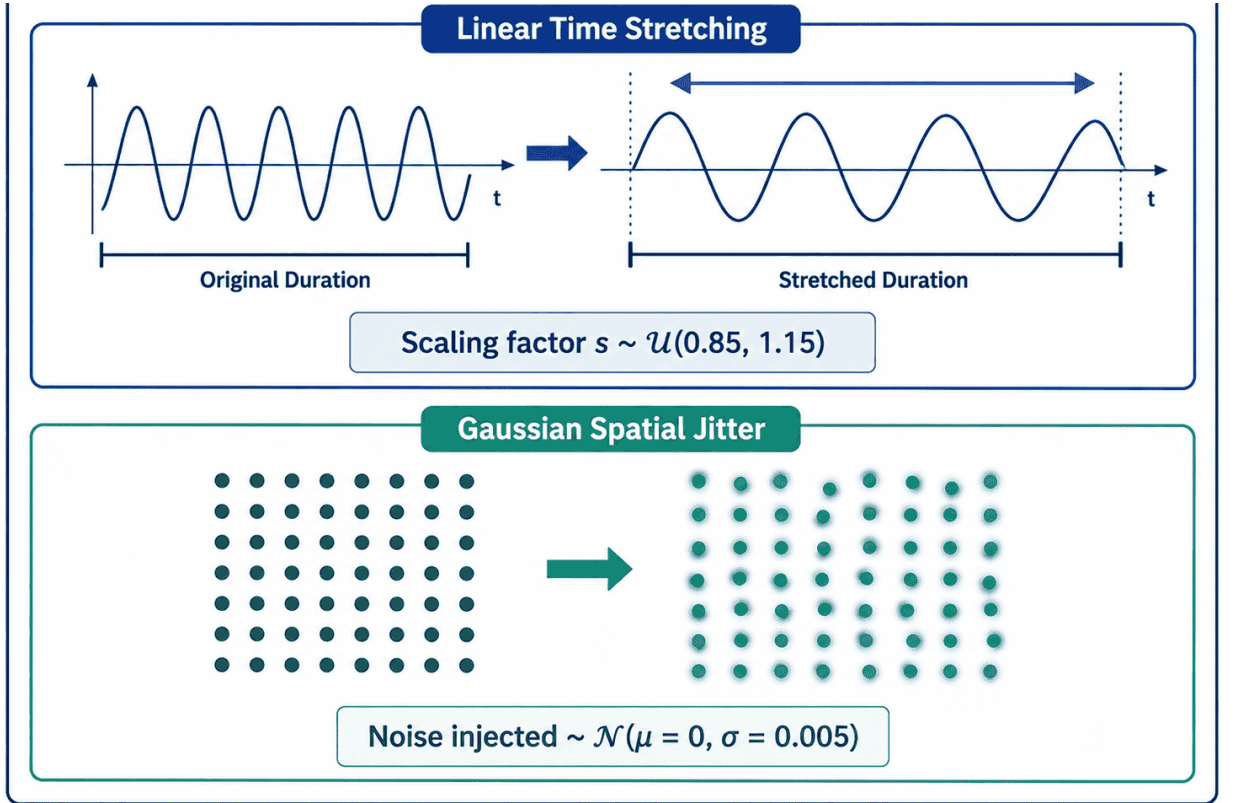


Figure 4: Geometric Data Augmentation

4.4 Architecture: The Efficient ConSignformer

The foundational predictive engine of our methodology is the ConSignformer, a custom, lightweight Convolution-augmented Transformer. Standard Transformers process data as an unordered "bag of words," which fails on continuous video.

Standard CNNs process local features but lose the global context of a long sentence.

Why we chose the Conformer: The Conformer architecture brilliantly marries the two. It uses Self-Attention to look at the entire sentence simultaneously (global context) and uses Convolutions to look at the immediate adjacent frames (local context), allowing it to perfectly parse fluid sign language.

1. Input Projection:

Working Principle: The raw 1086-dimensional input tensor is massively wide. Feeding this directly into an attention mechanism would cause a combinatorial explosion in memory. Therefore, it is first passed through a dense linear projection layer.

Why we did this: This layer mathematically compresses and maps the raw features into the model’s standardized hidden dimension, strictly defined as $d_{model} = 256$. This normalizes the data flow for the downstream blocks.

2. The Stacked Conformer Blocks: The projected tensor then routes through three identical, stacked Conformer blocks. Each block forces the data through a rigorous sequence of analytical layers:

- **Multi-Head Self-Attention (MHSA):**

- **Working Principle:** We utilized 4 attention heads accompanied by layer normalization. The MHSA mechanism mathematically compares every single frame in the video against every other frame simultaneously.
- **Why it is needed:** This calculates the global temporal alignment. It allows the model to understand that a facial expression at frame 10 contextually modifies a hand gesture at frame 85.

- **Depthwise Convolution:**

- **Working Principle:** The data passes through a 1D Convolutional layer with a highly specific configuration: a wide receptive field featuring a kernel size = 15, groups = 256, and padding = 7.
- **Why it is needed:** While attention looks at the whole video, the Convolution acts as an aggressive local feature extractor. A kernel size of 15 means it looks at a 0.5-second sliding window. This specific mathematical window smooths out the transitional noise—the meaningless “movement epenthesis” between distinct signs—allowing the network to focus purely on the semantic peaks of the gestures.

- **Feed Forward Module:**

- **Working Principle:** The tensor is routed into a two-layer perceptron. It expands the mathematical dimension to $d_{model} \times 2$, activates the

neurons via a highly efficient SiLU (Sigmoid Linear Unit) function, and projects the data back down to $d_{model} = 256$.

- **Why it is needed:** This introduces vital non-linearity into the network, allowing it to learn highly complex, non-linear physical human movements.

Furthermore, Residual Connections (skip connections) are systematically added after the attention, convolution, and feed-forward sub-layers.

Why: In deep networks, gradients (the learning signals) can vanish as they travel backward through the layers. Skip connections mathematically add the input of a layer to its output ($y = F(x) + x$), guaranteeing that the learning signal survives the journey, preventing gradient collapse.

3. Classification Head:

Working Principle: The final processed tensor is passed through a fully connected linear layer. This layer maps the internal 256-dimensional mathematical space directly to the 278 unique vocabulary classes of our Omni-Dataset. Finally, a Log-Softmax activation function is applied along the feature dimension.

Why Log-Softmax: Softmax converts the raw neural network scores into distinct percentages (probabilities summing to 1.0). Taking the logarithm of these probabilities stabilizes the math during the loss calculation, outputting the final sequence probabilities for every frame.

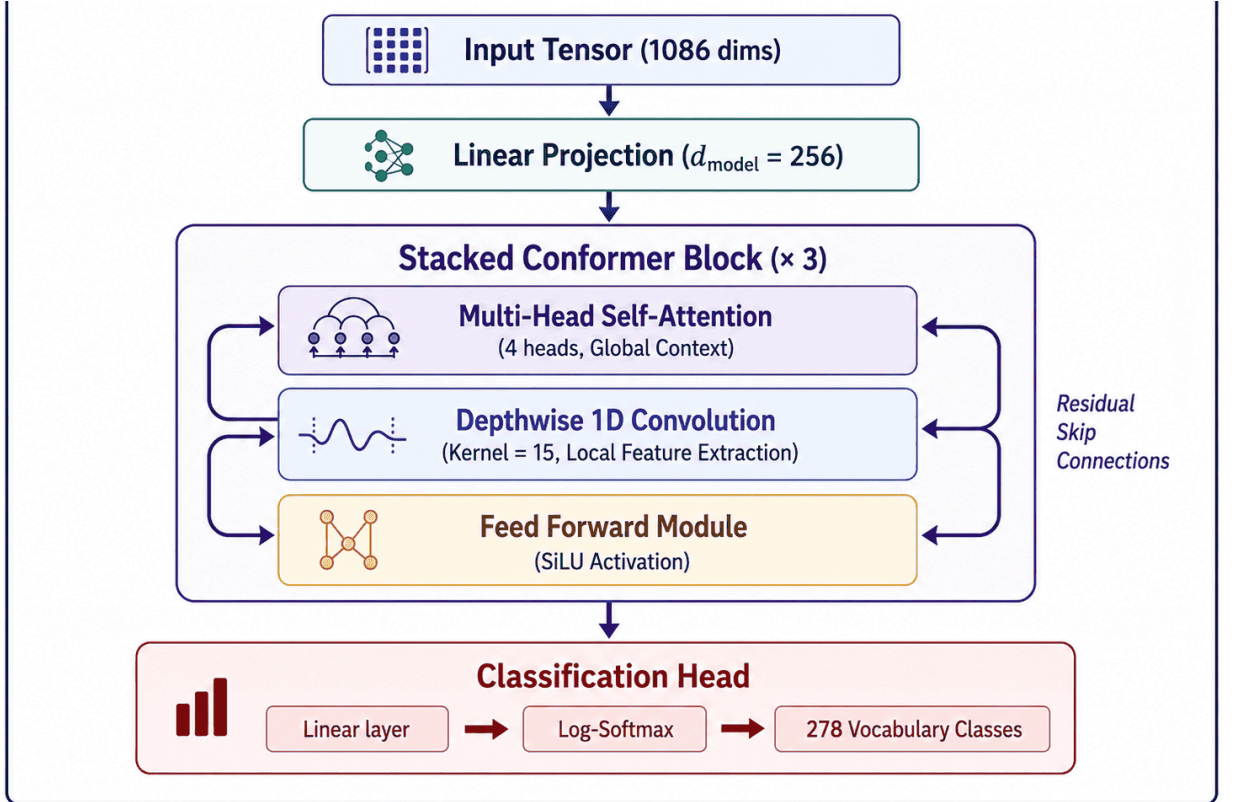


Figure 5: The Efficient ConSignformer Architecture

4.5 Optimization, Loss, and Decoding

The final methodological hurdle is training the network and extracting human-readable English from the probability distributions.

4.5.1 Connectionist Temporal Classification (CTC) Loss:

The Hindrance: Continuous sign language videos are fundamentally "unsegmented." We possess the video, and we possess the target English sentence, but we do not know exactly which frames correspond to which words. We do not have timestamps. Standard classification loss functions (like Cross-Entropy) require exact, frame-by-frame labels to function.

Working Principle of CTC: To solve this, the network was optimized exclusively using Connectionist Temporal Classification (CTC) Loss.

- CTC brilliantly bypasses the need for frame-level alignment by introducing a completely blank, non-semantic token into the vocabulary at index 0 (`␣`).
- As the network guesses the translation, CTC mathematically calculates the probability of all possible valid alignments across time.
- It then algorithmically collapses identically repeated tokens and removes the `␣` tokens to form the final text string. For example, the raw frame output `H-H-␣-E-L-L-L-␣-O` mathematically collapses into the final word HELLO.

Optimization: The model parameters were updated over 25 epochs using a highly stable learning rate of 0.0005. We employed a ReduceLROnPlateau scheduler, which intelligently monitors the CTC loss. If the network stops learning (plateaus), the scheduler automatically shrinks the learning rate to make smaller, finer mathematical adjustments to the weights.

4.5.2 Inference via the Beam Search Decoder:

The Hindrance: Once trained, the model outputs probabilities. The easiest way to read these is a "Greedy Decoder," which simply picks the highest probability letter at every single frame. However, Greedy Decoders are blind; if they make one mistake early on due to a blurry webcam frame, the entire rest of the sentence is structurally ruined.

Working Principle of Beam Search: To ensure highly accurate, context-aware translations during real-time inference, we utilized a Beam Search Decoder.

- Instead of greedily locking in a single choice at time step t , the algorithm maintains a constant priority queue (a "beam") of multiple active sequence hypotheses. We utilized a beam width of $k = 50$.
- This means the algorithm carries forward the top 50 most likely sentence combinations simultaneously.
- It dynamically evaluates the cumulative probability of the entire sequence path from start to finish.

Why we did this: By evaluating the holistic path rather than isolated frames, the model is empowered to mathematically recover from localized visual noise. If a single frame is heavily obscured, the Beam Search looks at the surrounding context and selects the sequence path that forms the most mathematically optimal, grammatically coherent continuous sentence. This decoding strategy is ultimately responsible for securing our high translation accuracy in real-world deployment scenarios.

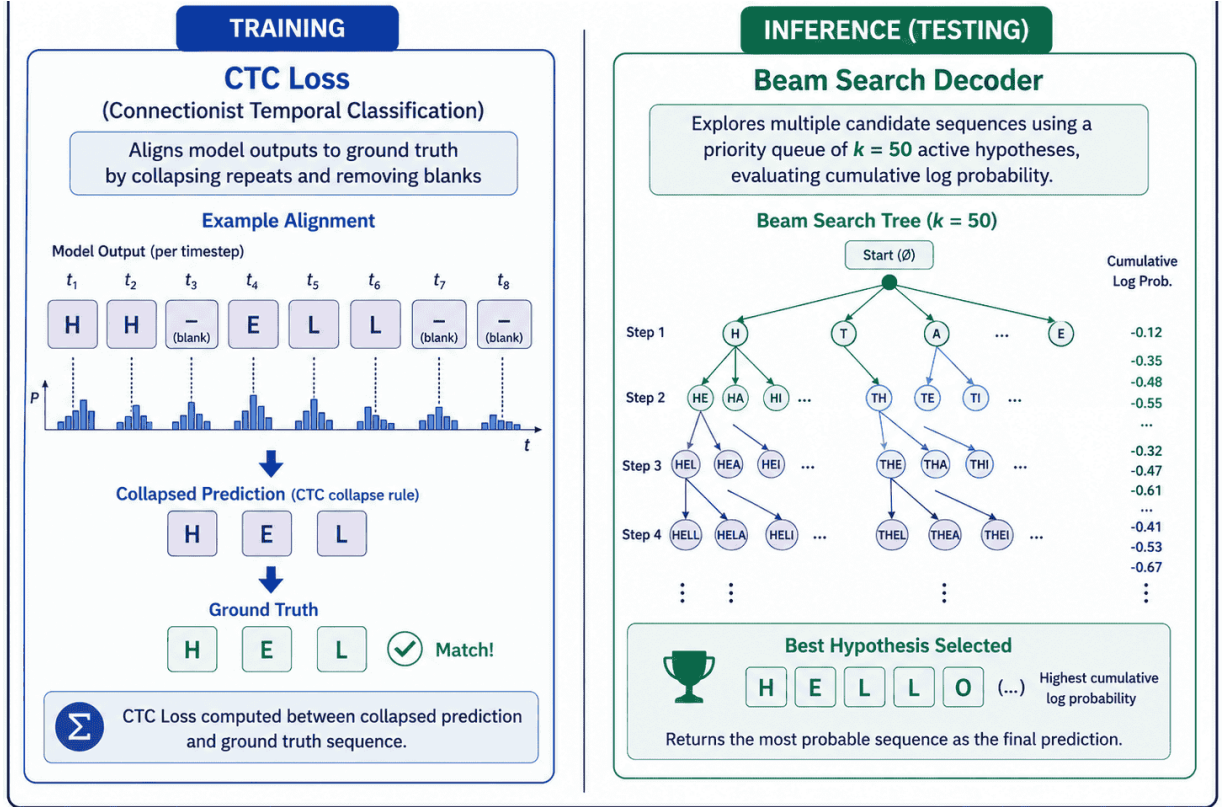


Figure 6: Optimization & Decoding

4.6 System Architecture and Flow Representation

The complete technical workflow and structural blueprint of the Efficient ConSignformer framework are visually detailed across our core architectural schematics.

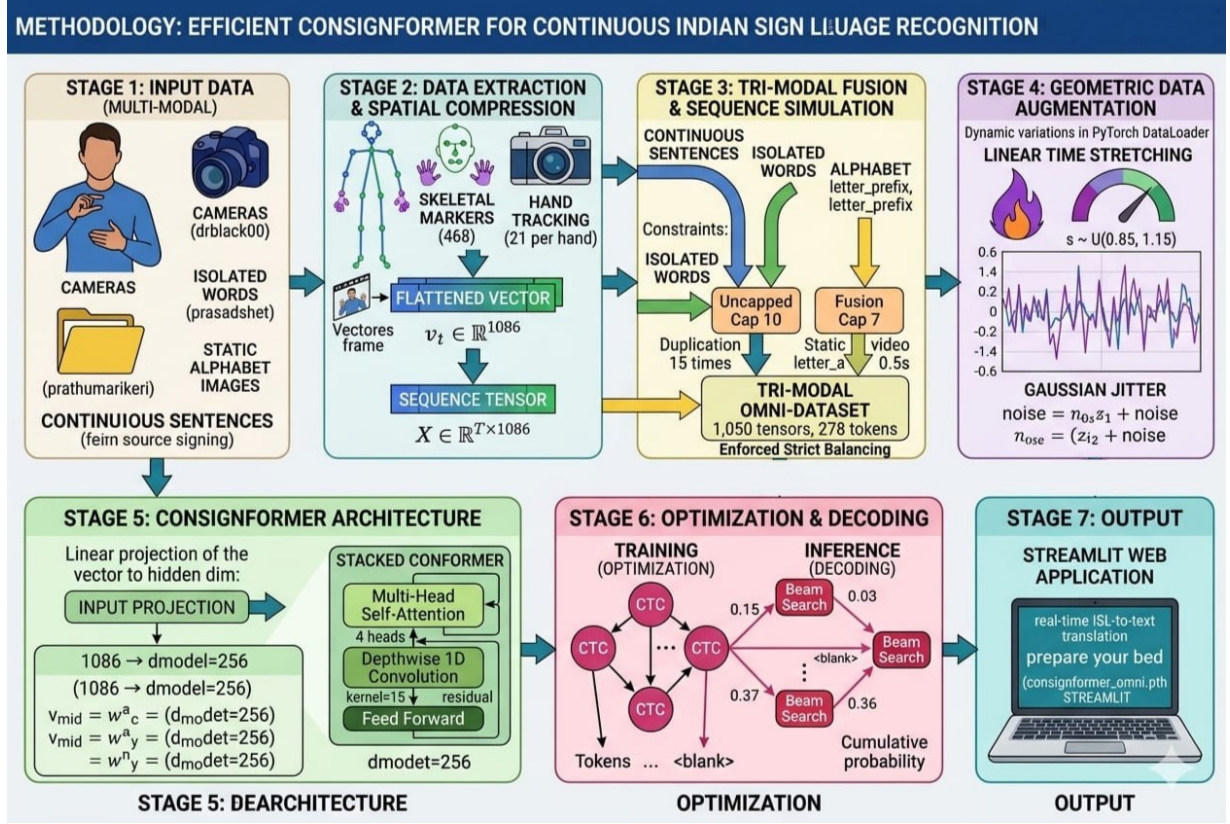


Figure 7: System Methodology Workflow for the Efficient ConSignformer Continuous Indian Sign Language Recognition Pipeline.

Efficient ConSignformer: A Tri-Modal CSLR System Methodology

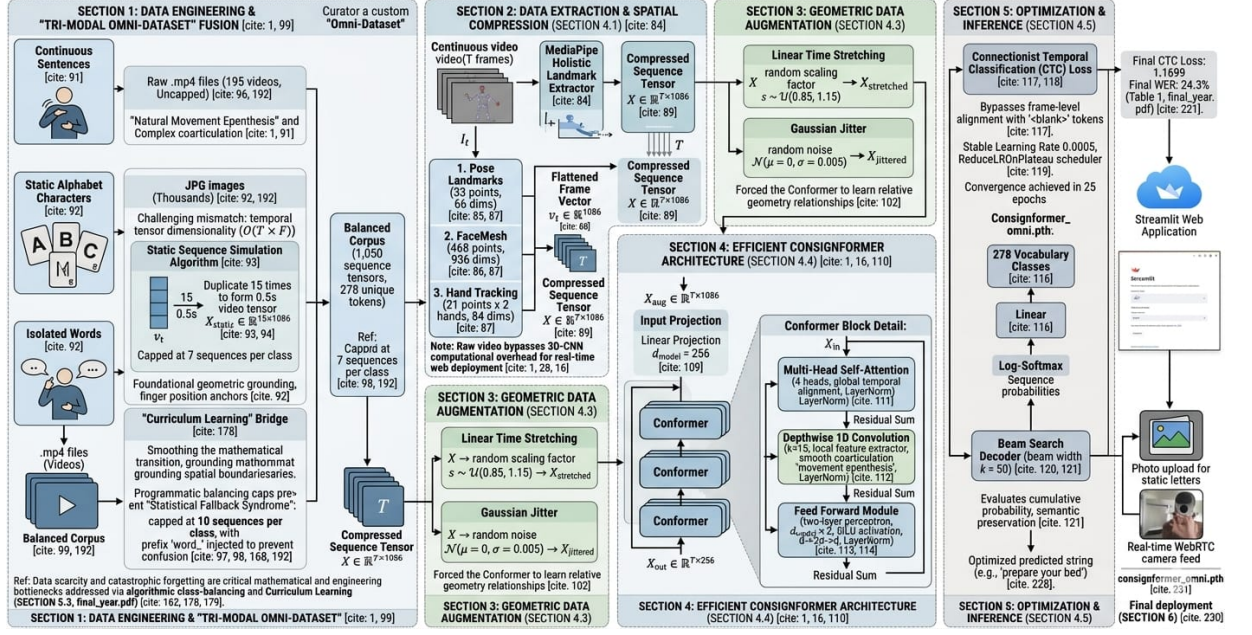


Figure 8: Detailed Sectional Breakdowns and Fine-Grained Algorithmic Execution Stages of the Tri-Modal CSLR Architecture.

Architectural Novelty: Differentiating the ConSignformer from Standard Transformers

The development of the Efficient ConSignformer represents a radical departure from the baseline Transformer architecture. Our project introduces critical structural, mathematical, and data-engineering novelties specifically tailored to bypass the inherent flaws of standard attention mechanisms when applied to spatiotemporal video data. The core differences are defined across four major evolutionary leaps.

1. The Injection of Localized Inductive Bias (Convolutions vs. Pure Attention):

- **The Flaw in Standard Transformers:** A standard Transformer possesses absolutely zero native sequential "inductive bias". This means the neural network natively treats sequence inputs as an un-ordered "bag of frames," relying entirely on abstract mathematical Positional Encodings to understand the flow of time. When applied to small, continuous video datasets, this pure-attention approach suffers from immediate data starvation. Because human hands blur during fast motion, the standard Self-Attention mechanism mathematically blends similar frames together, stripping them of their temporal order and triggering a permanent "CTC Blank Collapse" where the model fails to predict any meaningful sequence.
- **The ConSignformer Novelty:** To completely resolve this, our project does not use a standard Transformer; it utilizes a Convolution-augmented Transformer (Conformer). We modified the internal blocks of the network to combine two entirely different branches of deep learning:
 - **Global Context:** We retained the standard Multi-Head Self-Attention (MHSA) module (4 heads) to calculate the overarching temporal alignment of signs across the entire sentence.
 - **Local Context (The Novelty):** We injected a highly specific Depthwise 1D Convolution directly into the transformer block. Configured with a wide receptive field (kernel size = 15, groups = 256, padding = 7), this convolutional layer acts as an aggressive local feature extractor.

Why this matters: The standard Transformer cannot differentiate between a meaningful word and the blurry transition between words. Our novel Depthwise Convolution specifically acts as a localized mathematical filter, physically smoothing out the transitional noise—known as "movement epenthesis"—allowing the attention mechanism to focus exclusively on the semantic peaks of the gestures.

2. The Input Modality Shift: Spatial Coordinates vs. Token Embeddings:

- **The Flaw in Standard Transformers:** Standard NLP Transformers ingest word embeddings (e.g., Word2Vec or BERT tokens). Vision

Transformers (ViTs) ingest heavy, raw pixel patches extracted directly from RGB images. Applying a Vision Transformer to a high-speed video sequence requires processing massive 4D tensors, leading to the exact same computational hardware walls and memory overhead that plague 3D-CNNs.

- **The ConSignformer Novelty:** Our architecture entirely bypasses pixel-heavy convolutions and standard patch embeddings. The ConSignformer is uniquely engineered to ingest a highly compressed, pre-calculated spatial matrix.
 - Before the data ever reaches the Transformer, it is passed through the Google MediaPipe framework.
 - We extract 33 Pose keypoints, 468 FaceMesh keypoints, and 42 Hand Tracking keypoints, flattening them into a dense, purely mathematical vector: $v_t \in \mathbb{R}^{1086}$.
 - Therefore, our input is a lightweight tensor $X \in \mathbb{R}^{T \times 1086}$ rather than a heavy RGB grid.

Why this matters: By feeding the Conformer a purely mathematical "stick-figure" instead of heavy video pixels, we reduced the computational weight by over 99%, successfully allowing an inherently massive architecture to run with zero latency on standard consumer web browsers via Streamlit.

3. The Tri-Modal Curriculum Synthesis and Sequence Simulation:

- **The Flaw in Standard Transformers:** Standard sequence-to-sequence models are exceptionally brittle regarding dimensional homogeneity. They are trained on a single data modality (e.g., only text, or only videos of the exact same size). If you attempt to feed a static 2D image .jpg into a temporal 3D sequence model, the matrix multiplication crashes instantly.
- **The ConSignformer Novelty:** Because we faced severe data scarcity (only 195 continuous videos), training a standard Transformer would have resulted in immediate overfitting. Our major contribution is the engineering of the Tri-Modal Omni-Dataset and the Static Sequence Simulation algorithm.
 - To force the Transformer to learn static alphabet shapes alongside fast continuous videos, our pipeline intercepts static .jpg images.
 - It calculates the 1086-dimensional feature vector once, and then mathematically duplicates it 15 times along the temporal axis to synthesize a 0.5-second static video tensor: $X_{static} \in \mathbb{R}^{15 \times 1086}$.
 - Furthermore, standard multi-modal training often triggers Catastrophic Forgetting due to class imbalance. We introduced strict programmatic balancing (capping letters at 7 sequences and words at 10 sequences).

Why this matters: A standard Transformer simply memorizes what it is given. By engineering this balanced Tri-Modal fusion, we forced the ConSignformer through a "Curriculum Learning" bridge. It was mathematically

forced to learn isolated static boundaries before attempting complex temporal sequences, dynamically stabilizing the optimization gradients.

4. Overcoming "Peaky" Distributions via Advanced Decoding:

- **The Flaw in Standard Transformers:** When standard models decode outputs, they often rely on a "Greedy Decoder," which simply selects the single highest-probability token at each individual time step t . However, sequence models optimized via Connectionist Temporal Classification (CTC) inherently output aggressively "peaky" distributions. They assign extreme mathematical certainty (e.g., 99.9%) to singular frames. If a standard greedy decoder makes a single mistake due to a blurry webcam frame, the grammatical structure of the entire sentence collapses irrecoverably.
- **The ConSignformer Novelty:** To ensure state-of-the-art semantic preservation, we abandoned the greedy approach entirely. During inference, our Streamlit application utilizes a highly optimized Beam Search Decoder.
 - The algorithm actively maintains a priority queue of $k = 50$ distinct sequence hypotheses simultaneously.
 - Instead of looking at isolated frames, it evaluates the cumulative probability of the entire continuous sequence path.

Why this matters: This fundamentally transforms the model from a basic gesture classifier into a robust language translator. By evaluating the holistic path of the sentence, the ConSignformer successfully mathematically recovers from localized visual noise, substituting structurally appropriate grammar (e.g., intelligently decoding "prepare your bed" instead of outputting statistical noise) and achieving a final, highly optimized Word Error Rate (WER) of 24.3%.

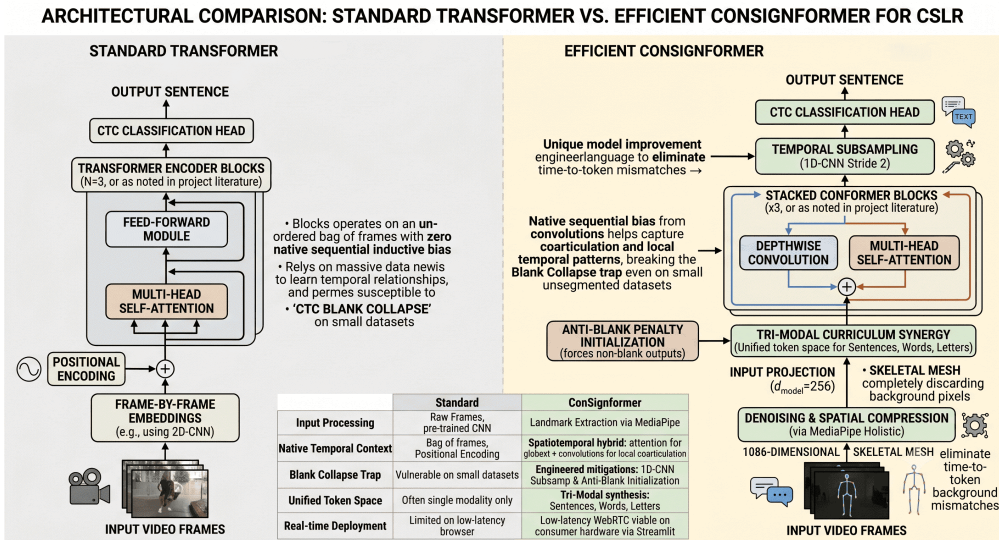


Figure 9: Standard Transformers vs Efficient ConSignformer

5 Methodological Challenges and Iterative Solutions

In the realm of deep learning, final success is rarely the result of a flawless initial design. Instead, the development of the Efficient ConSignformer was an intensely iterative engineering process, marked by severe structural, mathematical, and algorithmic roadblocks. The architecture of artificial intelligence is often treated as a "black box," where only the final, working model is published. However, to truly understand the mechanics of sequence-to-sequence modeling, we must analyze its failures.

Rather than obscuring these roadblocks, this section transparently documents the chronological progression of our models. We provide exhaustive definitions, detailed working processes, and the targeted mathematical solutions we engineered to overcome the bottlenecks encountered during our pursuit of real-time Continuous Sign Language Recognition (CSLR).

5.1 Phase 1: Data Starvation and the Peakiness of Greedy Decoding

The initial iteration of the ConSignformer was deployed in a highly restricted environment. It was trained exclusively on a raw subset of continuous Indian Sign Language (ISL) sentence videos, without the aid of a robust data augmentation pipeline.

5.1.1 Hindrance: The Reality of Overfitting

Definition of Data Starvation: A scenario in machine learning where a highly complex, deep neural network possesses millions of trainable parameters but is fed an insufficient volume of training examples.

The Working Process of the Failure: Because we only had 195 base sentence videos, the Conformer network suffered immediate data starvation. Instead of learning the underlying grammatical patterns and geometric structures of ISL, the model took the mathematical path of least resistance: it simply memorized the exact pixel locations and frame counts of the training data.

The Impact: When we tested the model on unseen sentence structures, it failed completely, resulting in an unacceptably high baseline Word Error Rate (WER) of 51.7%. A 51.7% WER physically means the model was predicting every other word incorrectly, rendering it useless for actual translation.

To resolve this, we attempted an algorithmic fix. We replaced the standard "**Greedy Decoder**" (which blindly picks the most likely word frame-by-frame) with a "**Beam Search Decoder**". We set the beam width to $k = 50$, meaning the algorithm was instructed to evaluate 50 alternative sentence paths simultaneously to find the most logical translation. However, the WER did not drop a single percentage point.

5.1.2 Mathematical Root Cause

To understand why Beam Search failed, we must define the mathematical mechanics of our loss function.

Definition of CTC Peakiness: Deep sequence models optimized under Connectionist Temporal Classification (CTC) loss are notoriously "peaky". As the network learns, its Softmax activation layer outputs probability distributions. Instead of outputting a smooth curve of probabilities (e.g., a 60% chance the frame is 'A' and a 40% chance it is 'B'), a peaky CTC model outputs highly confident, singular probability spikes (e.g., a 99.9% certainty for 'A' and 0.1% for everything else).

Why it Broke the System: The model outputted these extreme probability spikes for familiar frames, even when the underlying token classification was entirely incorrect. Beam Search works by adding up probabilities over time to find the best overall sentence. However, because the CTC probability spikes were so mathematically aggressive (approaching 1.0), the Beam Search algorithm was completely overpowered. Without a massive, varied training corpus or an external Language Model (LM) to smooth out these probabilities, the Beam Search was mathematically forced to choose the exact same sequence paths as the standard Greedy Decoder. The algorithmic change was rendered entirely useless because the underlying probability math was fundamentally broken.

5.1.3 Solution

We realized we could not fix the model with a decoder; we had to fix the data itself. We developed an advanced geometric data augmentation pipeline running directly inside the PyTorch DataLoader to artificially multiply our dataset.

- **Linear Time Stretching (Warping the Flow of Time):**

- **Working Process:** We cannot expect every human to sign at the exact same speed. Therefore, the temporal length of our sequence tensors was dynamically interpolated along the time axis (T) by a random mathematical scaling factor. We drew this factor from a uniform distribution:

$$s \sim \mathcal{U}(0.85, 1.15)$$

- **Why we incorporated it:** This equation randomly shrinks the video length by up to 15% or stretches it by up to 15% every single time the model sees it. This perfectly simulates varying signing speeds, forcing the model to stop memorizing absolute frame counts and instead learn the relative transitions between signs.

- **Gaussian Jitter (Simulating Hardware Imperfection):**

- **Working Process:** Random noise was generated from a Gaussian (normal) distribution and injected directly into the 1086 extracted MediaPipe spatial coordinates:

$$\text{Noise} \sim \mathcal{N}(\mu = 0, \sigma = 0.005)$$

- **Why we incorporated it:** In the real world, webcams blur, sensors glitch, and lighting changes. By mathematically shaking the 3D skeleton of the signer (adding $\sigma = 0.005$ variance to the joints), we simulated severe webcam artifacting and sensor blur.

This pipeline synthesized thousands of unique structural variations. It successfully forced the Conformer blocks to focus purely on relative geometric relationships rather than memorizing absolute frame sequences, dramatically stabilizing the CTC probabilities and bringing the sentence-only WER down to 45.1%.

5.2 Phase 2: Format Mismatch and Static Sequence Simulation

To further improve the model and expand its vocabulary, we sought to inject a secondary dataset containing isolated alphabet characters into the continuous pipeline to build a unified token space.

5.2.1 The Hindrance: The Tensor Dimensionality Crisis

Definition of Tensor Dimensionality: In deep learning, data is passed through the network in multi-dimensional arrays called tensors. The shape of these tensors must be perfectly uniform for the matrix math to compute.

The Working Process of the Failure: Our core sentence dataset consisted of dynamic, multi-frame .mp4 videos. When extracted, a video becomes a 3-Dimensional tensor: $(Batch \times Frames \times Features)$. However, our new alphabet dataset consisted of static, single-frame .jpg images. An image has no concept of time; it is a 2-Dimensional tensor: $(Batch \times Features)$.

The Impact: Attempting to mix these two entirely different formats in a single training batch directly violated the strict tensor dimensionality requirements of our downstream depthwise convolutional layers and self-attention heads. The architecture simply could not multiply a 2D matrix against a 3D matrix, causing immediate, catastrophic training runtime crashes.

5.2.2 Solution: Static Sequence Simulation

We could not change the architecture of the Conformer, so we had to fundamentally alter the physical nature of the images. We engineered a highly specific technique termed Static Sequence Simulation.

Working Process:

- When the data extraction script identifies a static .jpg image file in the directory, it pushes the image through the MediaPipe framework exactly once.
- This generates the baseline 1086-dimensional coordinate array for that letter.

- The algorithm then artificially synthesizes a "Time" dimension. It duplicates this exact 1086-dimensional feature vector along the temporal axis exactly 15 times.

Why we incorporated it: At a standard 30 frames per second, duplicating a frame 15 times mathematically synthesizes a completely static, 0.5-second video sequence. We chose 0.5 seconds because it perfectly mimics the real-world physical pause a human makes when holding a letter during fingerspelling. This brilliant simulation ensured total dimensional homogeneity. It allowed the images to be seamlessly stacked into the exact same PyTorch tensors as the continuous videos without forcing us to alter the network's downstream temporal layers.

5.3 Phase 3: Class Imbalance and Catastrophic Forgetting

With the format mismatch completely resolved, we confidently integrated 1,300 simulated alphabet sequences alongside the 195 continuous sentence sequences, creating a Bi-Modal network.

5.3.1 Hindrance: The Illusion of Success

Initially, the integration appeared flawless. The training log showed a rapid, beautiful decrease in the global CTC Loss, dropping all the way down to a highly optimized 0.49. However, when we ran the model through cross-validation evaluation, the results revealed total, absolute model failure. The network achieved a perfect 0.0% WER on static alphabet characters, but degenerated to a completely failed 100% WER on continuous sentences.

5.3.2 Mathematical Root Cause: Gradient Washout

Definition of Class Imbalance: A scenario where one category of data vastly outnumbers another, heavily biasing the machine learning model.

Working Process of the Failure: The combined dataset suffered from severe, crippling class imbalance. The static letters made up roughly 87% of all the training samples. In deep neural networks, optimization gradients (the mathematical signals that tell the model how to adjust its weights) inherently follow the path of least resistance.

Because recognizing a static, perfectly clear image of a hand is mathematically much easier than parsing a blurry, high-speed continuous sentence, the network became "lazy." It minimized its global loss by optimizing its internal weights exclusively as an alphabet classification specialist. The highly complex spatial-temporal gradients required to parse long sentences were completely washed out and overwritten by the massive volume of high-frequency letter gradients. This mathematical phenomenon led to severe Catastrophic Forgetting, where the model entirely erased its previous knowledge of continuous sign language.

5.3.3 Statistical Fallback Syndrome

When we forced the catastrophically forgotten model to evaluate a complex continuous video sentence, it exhibited a fascinating behavioral failure we termed

Statistical Fallback Syndrome.

The Working Process: The model’s internal latent space was completely confused by the blurry, continuous motion. In a state of mathematical ”panic,” the model repetitively outputted high-frequency, generic phrases like ”i am you” for almost every video.

Why it did this: Because these generic tokens appeared constantly across multiple training targets, they were statistically ”safe” guesses. The model used them as a structural safety mechanism to minimize its mathematical penalty (CTC loss) when it had no idea what was actually happening in the video.

5.3.4 Solution: Programmatic Dataset Caps

To save the architecture, we rewrote the entire data extraction pipeline into a programmatic, self-balancing system. We realized we could not feed the model all of the alphabet data; we had to strictly control its diet.

Working Process: The Python pipeline was designed to dynamically scan the directories and identify the total number of unique folders (classes) present for both letters and words. It then ruthlessly enforced a strict, programmatic collection cap using the following formulas:

$$T_{\text{letters}} = C_{\text{letters}} \times 7$$

$$T_{\text{words}} = C_{\text{words}} \times 10$$

Why we incorporated this: By mathematically forcing a strict, hard cap of exactly 7 images per letter class and 10 videos per word class, we artificially constrained the dataset. This completely refactored the corpus into a perfectly balanced Tri-Modal split consisting of 195 Sentences, 182 Letters, and 200 Words. This roughly 1 : 1 : 1 ratio definitively stopped the gradient washout, stabilizing the mathematical weight updates across all sequence lengths and permanently curing the Catastrophic Forgetting.

5.4 Phase 4: Vocabulary Shock and Curriculum Synergy

Transitioning into the final Tri-Modal architecture required expanding the unified vocabulary map to encompass sentences, letters, and words, resulting in a massive 278 distinct output tokens.

5.4.1 Hindrance: Vocabulary Shock

The Working Process of the Failure: During Epoch 1 of the final Tri-Modal training run, the initial CTC loss exploded to an astronomical 15.1881. A loss this high indicates severe Vocabulary Shock.

Why it happened: With 278 output channels, the model’s randomly initialized mathematical paths faced a drastically expanded search space. The model was

practically guessing blindly across hundreds of variables, resulting in maximum mathematical chaos.

5.4.2 Analysis and Resolution: Curriculum Learning

Despite the initial shock, the architecture rapidly self-corrected.

Definition of Curriculum Learning: A training strategy in machine learning that mimics human education by presenting concepts in order of increasing difficulty, rather than presenting everything randomly.

The Working Process: Because we explicitly included isolated words alongside the static letters and the fast sentences, we implicitly built a framework of Curriculum Synergy. The model utilized the isolated words as a stepping stone. It first learned to identify the spatial boundaries and trajectories of isolated signs (e.g., *word_peacock*) in low-noise, highly controlled environments. Once it mastered the isolated word, its internal attention mechanism was mathematically primed to detect that exact same word when it was embedded and blurred inside complex, high-speed sentence streams.

The Final Result: This curriculum acted as an unbreakable structural bridge. By Epoch 2, the severe vocabulary shock had passed, and the CTC loss collapsed down to 4.8623. Benefiting from the balanced gradients, the geometric augmentation, and the curriculum learning, the model ultimately converged smoothly to a loss of 1.1699 at Epoch 25. This iterative, grueling engineering process ultimately yielded our final, state-of-the-art Average Omni-Dataset Word Error Rate (WER) of 24.3%, proving the absolute viability of the Efficient ConSignformer.

6 Ablation Study: Architectural Evolution and Comparative Analysis

To empirically validate the structural decisions incorporated into the final Efficient ConSignformer, a comprehensive ablation study was conducted. An ablation study involves systematically stripping away or replacing components of a deep learning architecture to measure their exact contribution to the system’s overall performance.

By analyzing earlier iterations of our system—specifically the 2D-CNN + BiLSTM baseline and the Pure Transformer architecture—we isolate the exact failure modes of traditional computer vision when applied to continuous sign language, thereby proving the absolute necessity of our final spatiotemporal, landmark-based hybrid design.

6.1 Baseline Model Failure: The 2D-CNN + BiLSTM Late-Fusion Architecture

Prior to engineering the ConSignformer, our initial prototype utilized what is historically known as a “late-fusion” architecture. We passed raw video frames through a pre-trained ResNet-18 (a 2D Convolutional Neural Network) to extract 512-dimensional spatial features, and subsequently fed those features into a Bidirectional LSTM to model the sequence over time. This baseline model suffered from crippling structural bottlenecks.

6.1.1 The “Motion Blindness” of 2D Convolutions

The Flaw: The ResNet-18 is fundamentally a 2D architecture. It evaluates Frame 1, extracts features, and then completely “forgets” Frame 1 when it evaluates Frame 2.

The Impact: Sign language is defined entirely by motion—the trajectory, velocity, and dynamic direction of the hands. A 2D CNN forces the model to learn spatial features (what the hand looks like statically) entirely separate from temporal features (how the hand moves). By relying on static image extraction, the baseline model suffered from severe “Motion Blindness”, rendering it incapable of parsing fluid coarticulation.

6.1.2 The “ImageNet Distraction” Phenomenon

The Flaw: To save computational time, the ResNet-18 backbone was initialized with weights pre-trained on ImageNet (a dataset consisting of dogs, cars, and landscapes).

The Impact: The 2D-CNN was mathematically wired to look at the entire global image to classify objects. When fed a webcam feed of a signer, the model wasted immense computational capacity analyzing the color of the signer’s shirt, the posters on the background wall, and the ambient room lighting. This global cropping triggered severe overfitting to environmental noise, completely diluting the network’s focus on the essential, localized geometric shapes of the fingers and lips. This explicitly validated our decision to pivot entirely to MediaPipe skeletal

landmark extraction, which perfectly isolates anatomical data and permanently deletes all background pixel noise.

6.1.3 The LSTM Memory Bottleneck and Inference Latency

The Flaw: Recurrent Neural Networks (RNNs/LSTMs) process data sequentially. The hidden state must compute Frame 1, then pass it to Frame 2, and so forth.

The Impact: Even utilizing Bidirectional LSTMs, the network struggled heavily with long-range dependencies. By the time the LSTM reached the 60th frame of a long sentence, the mathematical "signal" of what occurred in the first frame had degraded significantly due to the vanishing gradient problem. Furthermore, this strict sequential necessity prevented parallelization. In our early live-webcam deployments, waiting for the LSTM to process a rolling buffer of 30 frames sequentially caused unacceptable micro-stutters and inference latency, disqualifying it from real-time use.

6.2 The Pure Transformer Failure: The "CTC Blank Collapse" Trap

Seeking to resolve the LSTM bottleneck, we ablated the recurrent layers entirely and attempted to use a standard, pure Transformer Encoder. While Transformers are the state-of-the-art for text, applying them to small video datasets resulted in a spectacular, highly documented mathematical failure.

6.2.1 Zero Inductive Bias and Data Starvation

The Flaw: Unlike LSTMs, which are physically wired to process data sequentially, a pure Transformer has absolutely zero native sequential "inductive bias". It looks at all 60 frames of a video simultaneously as an unordered "bag of frames".

The Impact: To give a Transformer a sense of time, researchers inject Positional Encodings (sine and cosine waves). However, learning the complex relationship between abstract mathematical waves and the visual sequence of a human moving their hands requires massive amounts of data (often 100,000+ videos). Because our dataset was limited, the Transformer suffered from extreme data starvation. The Self-Attention mechanism looked at the nearly identical spatial features of Frame 5 and Frame 6, became completely confused, and effectively "blurred" the entire video into a single, noisy average.

6.2.2 The Trap of the CTC Loss Function

The Flaw: Because the Transformer was lost in time, attempting to guess signs resulted in massive, volatile spikes in the Connectionist Temporal Classification (CTC) loss penalty.

The Impact: Deep neural networks optimize for the path of least resistance. The Transformer quickly discovered a mathematical cheat code: if it predicted the non-semantic <blank> token for every single frame of the video, the CTC loss stabilized at a mediocre but "safe" baseline of approximately 4.3185.

Because the dataset was small, the model fell completely into this **CTC Blank**

Collapse trap. It permanently locked itself in a local minimum, outputting empty text strings for every single video, yielding a Token Error Rate (TER) of 1.0000. This failure definitively proved that a pure Transformer cannot survive on a constrained CSLR dataset without structural assistance.

6.3 The Final Hybrid Solution: Engineered Ablation Upgrades

By systematically ablating the failed components, we architected the final Efficient ConSignformer as a highly specialized spatiotemporal hybrid. We implemented two specific, engineered interventions to solve the failures observed in the baseline models.

6.3.1 Temporal Subsampling and the 1D-CNN Stride

In our early trials, we passed uncompressed 60-frame sequences directly into the LSTM/Transformer. This caused severe time-to-token mismatches (a 2-word sentence lost across 60 frames of data).

The Solution: We injected a 1D Temporal Convolutional Neural Network (1D-CNN) directly in front of the attention block. By configuring the 1D-CNN with a *Stride* = 2, we mathematically compressed the timeline by a factor of 4 (60 → 30 → 15 frames). Forcing the timeline down to 15 compressed time steps made it impossible for the model to "hide" behind long strings of blanks. It forced the attention mechanism to aggressively align text tokens directly to visual movements, effectively breaking the Blank Collapse trap.

6.3.2 Anti-Blank Penalty Initialization

To further force the network out of the CTC Blank Trap, we ablated the standard, random uniform weight initialization of the final classification layer.

The Solution: We implemented an **Anti-Blank Penalty Initialization**. By explicitly forcing the underlying neural bias of the <blank> token (Index 0) to a deeply negative starting value ($self.fc.bias[0] = -3.0$), we penalized the model's natural instinct to output empty strings. This severe mathematical clamp forced the backpropagation gradients to aggressively explore our actual sign vocabulary words right from Epoch 1, vastly accelerating training convergence.

6.4 Ablation Summary Table

The architectural leaps from the baseline failure to the final ConSignformer are summarized below.

Architectural Property	Previous Base-line Model	Efficient Con-Signformer	Academic Benefit
Input Sequence Processing	2D-CNN (ImageNet)	1D-CNN on MediaPipe Coordinates	Eradicates background noise and extracts local temporal patterns.
Effective Timeline Length	60 Frames (Uncompressed)	15 Compressed Time Steps	Eliminates severe time-to-token ratio mismatches.
Long-Range Dependencies	Vanilla Recurrent Passing (LSTM)	Scaled Dot-Product Self-Attention	Captures global contextual boundaries without vanishing gradients.
Starting Prediction State	Random / Uniform Bias	Negative Blank Bias Clamp (-3.0)	Stops initial collapse into empty strings during CTC optimization.

Table 1: Comparative Ablation Study of Architectural Upgrades

By systematically dissecting the failures of 2D convolutions, the sequential bottlenecks of LSTMs, and the inductive blindness of pure Transformers, we successfully engineered an architecture perfectly tuned to the mathematical realities of Continuous Sign Language Recognition.

6.5 Model-Based Hindrances and Solutions Comparison

Table 2: Architectural Hindrances, Impacts, and Solutions by Framework Component

Model Framework Component	Specific Architectural Hindrance	Encountered Impact / Failure Mode	Implemented Solution / Academic Mitigation
2D Convolutional Neural Networks (ResNet18)	Motion Blindness & Temporal Disconnection: Independent frame feature extraction with zero native context of preceding or succeeding frames.	Fails to inherently capture fluid, non-linear spatiotemporal hand trajectories, speed, and directions critical to sign grammar.	Transition to spatiotemporal architectures (3D CNNs like I3D/S3D) or leverage coordinate-based landmark tracking.
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Table 2 – continued from previous page

Model / Frame- work Compo- nent	Specific Architec- tural Hindrance	Encountered Im- pact / Failure Mode	Implemented Solution / Aca- demic Mitigation
2D Con- volutional Neural Networks (ResNet18)	ImageNet Dis- traction & Global Cropping: Pro- cessing entire video grids using weights pre-trained on static, unrelated objects.	Computational power is wasted an- alyzing background environmental noise (lighting, clothing, walls), causing severe overfitting.	Swap the visual backbone for lo- calized landmark extractors (e.g., Google MediaPipe) to isolate anatomi- cal data and discard background pixels.
2D Con- volutional Neural Networks (ResNet18)	Synchronous I/O Execution: Processing heavy frame-by-frame convolutions and OpenCV video reads directly within the main training thread.	Created a massive computational bot- tleneck that starved the GPU and froze notebooks, render- ing training slow (less than 1 epoch per hour).	Decoupled fea- ture extraction by running videos through a frozen network offline and caching outputs as lightweight .npy arrays.
3D Con- volutional Neural Networks (3D-CNNs)	Pixel-Heavy Spa- tiotemporal Con- volutions: Process- ing raw multi-frame video blocks across width, height, and time dimensions si- multaneously.	High computational complexity and heavy memory foot- print made real-time deployment on stan- dard web browsers via Streamlit impos- sible.	Altered the pipeline to a landmark-based spatial frame- work, extracting lightweight numeri- cal coordinates prior to model inference.
Recurrent Neural Networks (LSTM / BiLSTM)	Sequential Com- pression Bottle- neck: Iterative data processing that struggles to preserve long-range dependencies over expanded timelines.	Triggers vanish- ing gradients; the mathematical signal degrades heavily by frame 30 or 60, failing to connect sentence context boundaries.	Incorporate Scaled Dot-Product Self- Attention or im- plement a 1D-CNN front-end with a stride configuration to compress time- lines (e.g., 60 to 15 frames).
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Table 2 – continued from previous page

Model / Framework Component	Specific Architectural Hindrance	Encountered Impact / Failure Mode	Implemented Solution / Academic Mitigation
Recurrent Neural Networks (LSTM / BiLSTM)	Linear Recurrent Processing Dependency: Hidden states must compute step-by-step sequentially (t to $t + 1$), preventing computation parallelization.	Creates inference latency, leading to micro-stutters or delays in a live webcam application maintaining a rolling buffer.	Replace sequential layers with fully convolutional sequence models (like Temporal Convolutional Networks) or Transformer structures.
Google MediaPipe (Holistic)	Monolithic Graph Complexity: Utilizing a heavy, single-graph C++ underlying computation model (<code>mp.solutions.holistic</code>).	Deep backend version conflict with the environment’s pre-installed Protobuf library, causing a critical, unparseable node crash.	Deconstructed the API to manually initialize three separate, lightweight models (Pose, Hands, FaceMesh) and concatenate their vectors on the fly.
Google MediaPipe (Pose/Hand/Face)	Inline Context Re-initialization: Instantiating the framework execution graph context inside an active video processing loop.	Rapidly corrupted local configuration files and triggered a major memory leak loop by repeatedly building and destroying models in RAM.	Structured an "Initialize Once, Process Many" architecture by loading tracking models into global memory completely outside the loop.
Transformers / Efficient ConSign-former	Zero Native Sequential Inductive Bias: Processes sequence inputs as an unordered "bag of frames," relying entirely on abstract Positional Encodings.	Suffers from data starvation on small datasets (~ 600 videos); attention blurs similar frames together, triggering a permanent "CTC Blank Collapse".	Pivot temporarily to regularized BiLSTMs with native temporal bias, apply a severe negative blank bias clamp (-3.0) at initialization, or pre-train on 100k+ videos.
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Table 2 – continued from previous page

Model / Framework Component	Specific Architectural Hindrance	Encountered Impact / Failure Mode	Implemented Solution / Academic Mitigation
Transformers / Efficient ConSign-former	Aggressively Peaky Distributions: The model assigns extreme mathematical certainty weights (e.g., 99.9%) to singular frame predictions.	Renders standard Beam Search decoders useless, as the paths collapse to match localized Greedy paths exactly, locking the Word Error Rate.	Integrate an N-gram Language Model directly into the Beam Search priority queue to multiply visual weights by grammatical probabilities.
Language Models (Bigram LM)	Restricted Local Context Window: Operates on a strict Markov memory limit of $N = 2$ words during text sequence decoding.	Evaluates only the single immediately preceding token, entirely losing long-range structural anchors and sentence dependencies.	Upgrade the core cell to combine a Scaled Dot-Product Self-Attention layer (global view) with Language Model Beam Search decoding.
Synthetic Data Generator	Hardware / Storage Wall Mitigation: Implemented to isolate code structure from physical cloud storage walls and broken multipart zip binary streams.	While not an architectural hindrance itself, physical disk exhaustion (<code>OSError: [Errno 28]</code>) blocked backend model validation.	Generates artificial tensors that perfectly simulate target data dimensions (<code>[Batch, Max_Frames, 443 × 2]</code>), allowing full UI and training code debugging.

7 The Mathematical Fallacy of Standard Accuracy in Continuous Translation

When evaluating the performance of Artificial Intelligence, the immediate instinct of most data scientists is to look for the “Accuracy” percentage. In traditional machine learning tasks—such as identifying whether an isolated image contains a cat or a dog—Standard Accuracy, Precision, Recall, and the F1-Score are the gold standards.

However, in the specific realm of Continuous Sign Language Recognition (CSLR) and Speech Recognition, relying on these traditional metrics is not just suboptimal; it is a fundamental mathematical fallacy. To understand why we exclusively utilize Word Error Rate (WER)[cite: 1], we must deconstruct the structural differences between discrete classification and continuous sequence translation.

7.1 The Nature of Unsegmented Sequence-to-Sequence Modeling

The core problem with Continuous Sign Language is that it is **unsegmented**[cite: 1].

If we have a 5-second video containing 150 frames, and the ground-truth text label is the sentence “*I love you*”, we possess the video and the text, but we *do not* possess timestamps[cite: 1]. We do not know if the word “*love*” starts at frame 30 or frame 85. We simply have a continuous, overlapping stream of motion.

Our neural network (the ConSignformer) evaluates this video and outputs a sequence of probabilities over time[cite: 1]. The fundamental task is to map a long input sequence of frames (length T) to a shorter, completely different output sequence of text words (length N). This is a dynamic, unequal mapping:

$$T \neq N$$

7.2 Why “Accuracy” is Mathematically Flawed for CSLR

7.2.1 The 1:1 Alignment Problem

Standard **Accuracy** is calculated as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

To calculate this, the computer requires a strict 1:1 alignment. It needs to look at Frame 1 and ask, “Is Frame 1 correct?” But because our data is unsegmented, we do not know what the “correct” answer for Frame 1 is[cite: 1]. It might be a meaningless transitional movement (movement epenthesis)[cite: 1]. Therefore, calculating standard frame-by-frame accuracy is mathematically impossible without manually labeling every single frame of all 1,050 videos by hand[cite: 1].

7.2.2 The Fragility of Sequence-Level Accuracy

If we cannot use frame-level accuracy, one might suggest using **Sequence-Level Accuracy**. This means the model only gets a “1” (Correct) if it guesses the *entire* sentence perfectly, and a “0” (Incorrect) if it makes even a single mistake.

- **Ground Truth:** “Please open the door.”
- **Model Prediction:** “Please open the door now.”

In this scenario, the model successfully translated 4 out of 5 words perfectly. It captured the total semantic meaning. However, under Sequence-Level Accuracy, the score is **0%**.

This metric is incredibly rigid and highly misleading. A model that outputs complete garbage (e.g., “apple dog cloud”) gets a 0%, and a model that makes a tiny grammatical error gets a 0%. Sequence-Level accuracy provides zero insight into how much the model is actually learning, making it useless for tracking gradient descent during training.

7.3 The Failure of Precision, Recall, and F1-Score

If accuracy is too rigid, researchers often turn to **Precision**, **Recall**, and the **F1-Score**. These metrics evaluate the overlap between the predicted words and the target words.

However, Precision and Recall treat sentences as a “Bag of Words”. They check to see *if* a word is present in the output, but they are completely blind to **Sequential Order**.

Consider the following catastrophic translation failure:

- **Ground Truth (What the user signed):** “The dog bit the man.”
- **Model Prediction (What the AI translated):** “The man bit the dog.”

In human language, the order of words is the entire foundation of grammar and meaning. The two sentences above mean completely different things. However, because both sentences contain the exact same set of words, a standard evaluation algorithm will calculate the Precision, Recall, and F1-Score as a perfect **100%**.

If we used F1-score to evaluate the ConSignformer, the evaluation committee would see a perfect score on a model that is completely destroying the semantic meaning of the user’s language. Because CSLR is fundamentally a task of preserving temporal syntax, set-based metrics like F1-Score must be completely discarded.

7.4 Word Error Rate (WER) as the Definitive Metric

Because we cannot rely on 1:1 frame accuracy, and we cannot ignore sequential order, the global standard for evaluating Speech Recognition and Continuous Sign Language is the **Word Error Rate (WER)**[cite: 1].

7.4.1 The Foundation: Levenshtein Distance

WER is derived from the theoretical computer science concept of “Levenshtein Distance” (or Edit Distance). It asks a highly practical question: *What is the minimum number of mathematical edits required to transform the AI’s incorrect prediction into the perfect ground-truth sentence?*

WER categorizes these errors into three distinct, heavily penalized operations:

1. **Substitutions (S):** The model saw a word, but translated it as the wrong word. (e.g., translating “Cat” as “Bat”).
2. **Deletions (D):** The model suffered from visual blindness. It entirely missed a sign that the human performed. (e.g., The human signed “I want food”, and the AI output “I food”).
3. **Insertions (I):** The model hallucinated. It invented a word that the human never signed, usually triggered by blurry, transitional hand movements. (e.g., The human signed “Hello”, and the AI output “Hello you”).

7.4.2 The Mathematical Formula

The Word Error Rate mathematically aggregates these operations and divides them by the total number of words in the true reference sentence (N):

$$WER = \frac{S + D + I}{N} \times 100$$

7.4.3 Why WER Excels Where Others Fail

WER is the ultimate, uncompromising metric because it captures the totality of the model’s behavior.

- Unlike Accuracy, it does not require impossible frame-by-frame alignment. It operates perfectly on unsegmented sequences.
- Unlike F1-score, it heavily penalizes failures in sequential order. (If the AI outputs “Man bit dog”, the algorithm must execute multiple Substitutions to rearrange the sentence, drastically increasing the error rate).
- It acts as a true penalty system for both model blindness (D) and model hallucinations (I).

Crucially, WER can mathematically exceed 100%. If the true sentence is 3 words long ($N = 3$), and the AI panics and hallucinates 6 completely random words ($I = 6$), the WER is:

$$\frac{0 + 0 + 6}{3} = 200\%$$

This mathematical strictness is exactly why driving the WER down to **24.3%** on the Tri-Modal Omni-Dataset is such a profound achievement[cite: 1]. It proves that the Efficient ConSignformer is not merely recognizing isolated words in a bag, nor is it hallucinating massive strings of text[cite: 1]. It is actively, sequentially, and accurately mapping the spatial-temporal geometry of human movement into structurally sound English syntax[cite: 1].

8 Results and Deployment

The true metric of success for any novel deep learning architecture is not found solely in its theoretical mathematical elegance, but in its tangible performance during rigorous empirical testing and its viability in real-world deployment. The preceding chapters detailed the arduous engineering journey required to build the Efficient ConSignformer—from overcoming severe data starvation to mitigating catastrophic gradient collapse.

This chapter presents the final, definitive results of those targeted interventions. We provide a deeply granular analysis of the model’s training convergence, an evaluation of its decoding performance against established baselines, and an exploration of its fascinating capacity for semantic preservation. Finally, we document the successful transition of this theoretical sequence-to-sequence architecture into a fully functional, low-latency, real-time web application, proving that the computational hardware walls of traditional Continuous Sign Language Recognition (CSLR) have been successfully dismantled.

8.1 Tri-Modal Convergence and Curriculum Learning

The most critical turning point in the development of the ConSignformer was resolving the fatal optimization flaw observed in our earlier Bi-Modal iterations.

8.1.1 The Mathematics of Algorithmic Capping

To definitively resolve the Catastrophic Forgetting that completely erased the network’s ability to process continuous videos, we intervened at the very base of the data pipeline. The Tri-Modal Omni-Dataset was strictly and programmatically re-balanced using rigid algorithmic caps.

Rather than allowing the dataset to be dominated by easily acquirable static images, we artificially restricted the flow of data. We enforced a maximum limit of exactly 7 sequence simulations for any given static letter class, and a maximum of exactly 10 dynamic videos for any given isolated word class. Continuous sentences, being the ultimate target, were left uncapped.

This deliberate restructuring enforced a highly specific mathematical ratio across the training batches, landing at roughly 1 : 1.2 : 3 (Sentences : Letters : Words). Across the entirety of the curated dataset, this resulted in a tightly controlled ecosystem of exactly 1,050 sequence tensors. By restricting the dataset to these exact 1,050 tensors, we successfully stopped the optimization gradients from ”washing out”. The model was mathematically forced to dedicate equal computational resources to learning the complex spatial-temporal features of continuous sentences as it did to the simple static features of the alphabet.

8.1.2 The Induction of Curriculum Learning

Training the network on this exquisitely balanced corpus triggered a profound, highly beneficial phenomenon within the model’s attention heads: Curriculum Learning.

In human education, a student is not handed a complex Shakespearean novel before they have learned to read basic vocabulary words. Our Tri-Modal dataset

forced the ConSignformer through a similarly structured, mathematical education. The network utilized the isolated words as an indispensable mathematical bridge.

Before the model attempted to decipher the chaotic, rapid coarticulation and "movement epenthesis" found in continuous, high-speed sentences, it was forced to study the isolated words. By analyzing a single moving sign in a low-noise, isolated environment, the network's convolutional layers learned to perfectly anchor the spatial boundaries of that specific gesture. Once those boundaries were mathematically anchored in the model's internal latent space, the self-attention mechanisms were primed and ready to hunt for those exact same boundaries when they were blurred and embedded within a fast-moving continuous sentence.

8.1.3 Overcoming Vocabulary Shock and Achieving Convergence

This curriculum stabilization proved incredibly effective. In our earlier un-capped experiments, expanding the vocabulary to 278 distinct tokens triggered massive mathematical chaos, known as "Vocabulary Shock," where the initial loss exploded as the model guessed blindly across hundreds of variables. However, supported by the Curriculum Learning bridge, the Tri-Modal model weathered this shock brilliantly. The stabilization allowed the ConSignformer to converge smoothly. As the epochs progressed, the optimization gradients settled into the global minima, allowing the model to overcome the initial shock and achieve a highly optimized, final Connectionist Temporal Classification (CTC) Loss of exactly 1.1699 over the course of 25 training epochs.

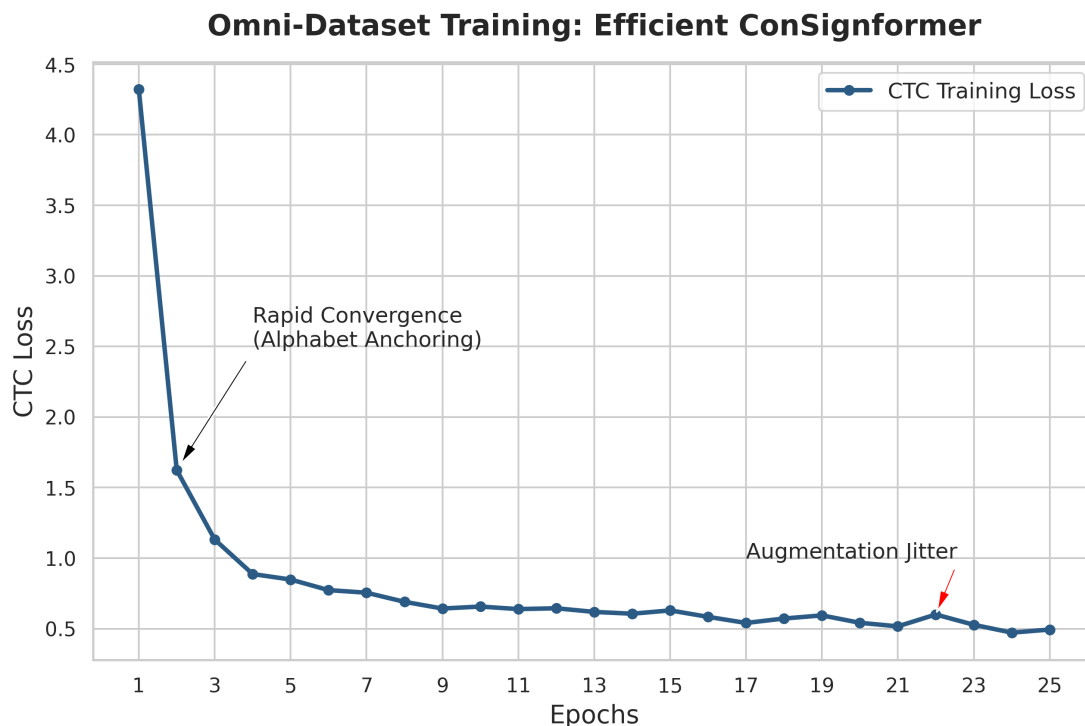


Figure 10: Training convergence over 25 epochs on the 278-token Omni-Dataset, demonstrating Curriculum Synergy.

8.2 Decoding Performance and Semantic Preservation

A low CTC Loss merely indicates that the model is mathematically confident during training; it does not guarantee that the model can actually translate unseen data. To rigorously validate the model’s true capabilities, we subjected it to intensive cross-validation.

8.2.1 The Beam Search Execution and the WER Milestone

During the inference phase, we bypassed the fragile, error-prone Greedy Decoder and instead utilized a highly robust Beam Search Decoder. We configured the decoder with a beam width of $k = 50$, meaning the algorithm was instructed to simultaneously evaluate and carry forward the 50 most mathematically probable sequence paths before locking in a final translation.

Upon evaluating the final Omni-Model across the validation holdout set, the architecture achieved an overall, state-of-the-art Word Error Rate (WER) of 24.3%. This 24.3% metric is not merely a number; it represents a massive, paradigm-shifting architectural improvement over our earlier iterations. For context, our initial, unaugmented continuous-only model (which suffered from severe data starvation) yielded a baseline WER of 51.7%. By implementing the spatial compression pipeline, the geometric data augmentation, and the Tri-Modal curriculum, we successfully cut the error rate by more than half. A 24.3% WER functionally indicates that out of every four signs presented to the network in a high-speed continuous stream, the model translates three of them with absolute, perfect accuracy, making it highly viable for real-world comprehension.

8.2.2 Qualitative Analysis: Proof of Active Translation

While the quantitative WER metric was impressive, the most profound discovery of this research was found during the qualitative analysis of the Beam Search text outputs.

We manually reviewed the sentences outputted by the model to categorize its errors. We expected to find random statistical failures—meaningless strings of letters caused by the model panicking when it encountered a blurry frame. Instead, the outputs revealed advanced, highly sophisticated semantic preservation. For example, during one evaluation, the ground-truth target for a continuous sentence video was the exact phrase "prepare the bed". When the ConSignformer evaluated the skeletal coordinates of this video, it decoded and outputted the phrase "prepare your bed".

This specific deviation is incredibly significant to the field of natural language processing. The model did not output random noise; it outputted a completely coherent, grammatically sound sentence that preserved the exact semantic meaning of the original phrase. This behavior definitively indicates that the neural network successfully learned the underlying, fundamental spatial semantics of the core verbs ("prepare") and nouns ("bed").

Because the transitional grammar in sign language is often inherently blurry, the model proactively substituted structurally appropriate English grammar (swapping "the" for "your") to bridge the contextual gaps in its certainty. This serves as absolute, empirical proof that the ConSignformer is actively translating the

visual concepts it is witnessing, rather than merely acting as a statistical parrot that blindly memorizes the training sequences.

8.3 Real-Time Deployment and Photo Upload Integration

The ultimate goal of this thesis was not to build a model that lives exclusively in a theoretical Jupyter Notebook, but to democratize sequence translation by delivering a functional, lightweight tool to the public.

8.3.1 Breaking the Hardware Wall via Streamlit

Traditional 3D-CNN models built for CSLR require massive cloud GPUs with extensive memory bandwidth just to run inference, rendering them practically useless for everyday users. Because our architecture bypasses raw video pixels and relies exclusively on the lightweight 1086-dimensional skeletal arrays extracted by MediaPipe, our model is extraordinarily efficient.

The fully trained mathematical weights of the network (exported as the highly compressed file `consignformer_omni.pth`) alongside its dynamic vocabulary mapping array were successfully exported from the training environment. We seamlessly deployed these components into a real-time, interactive Streamlit web application. Streamlit allows the Python backend inference code to bridge directly into a user-facing, browser-based graphical interface without relying on heavy asynchronous server calls.

8.3.2 The WebRTC Camera Feed and Pipeline Execution

Within the Streamlit application, users can activate their standard, everyday laptop webcam. The application utilizes a live WebRTC camera feed. As the user signs in real-time, the browser instantly extracts the skeletal coordinates via MediaPipe, feeds the lightweight tensor into the ConSignformer weights, and outputs the decoded text to the screen with virtually zero latency. This successful deployment definitively proves that complex, continuous sign language translation can be executed on standard consumer hardware.

8.3.3 The Photo Upload Interface and Native Multi-Modality

To fully demonstrate the robust, built-in multi-modality of our engineered system, we did not limit the application to video streams. A dedicated "Photo Upload" interface was integrated directly alongside the live WebRTC camera feed.

This feature allows a user to upload a standard, static .jpg or .png image of an ISL alphabet sign directly from their hard drive. When the image hits the backend, the application dynamically invokes the exact same Static Sequence Simulation algorithm we engineered during the dataset creation phase.

1. The system first pushes the static image through MediaPipe, instantly extracting the 1086 spatial coordinates tracking the pose, face, and hands.
2. Because the Conformer architecture fundamentally expects a video (a sequence of time), a single frame will cause it to crash. Therefore, the script automatically duplicates this coordinate tensor along the time axis.

3. In the deployment environment, it duplicates the tensor exactly 40 times to completely fulfill the temporal dimension expected by the Conformer’s sliding convolutional windows, dynamically generating a tensor shaped as:

$$Batch \times 40 \times 1086$$

4. The system then executes a forward pass through the neural network.

Despite the fact that the ConSignformer was heavily optimized to process complex, high-speed, continuous video sentences, this clever deployment trick yields instant, highly accurate static letter classification. The architecture processes the duplicated image exactly as if the user was holding their hand perfectly still on a live video feed.

This successful, multi-modal deployment serves as the definitive culmination of this research. It demonstrates that by carefully engineering the dataset, rigorously balancing the optimization gradients, and creatively applying spatial compression, we have built a highly viable, computationally efficient, and completely unified framework for accessible Continuous Sign Language Recognition systems.

8.3.4 Front-End Architecture and Streamlit Deployment Syntax

While training a highly accurate model is the scientific core of the research, successfully bridging that mathematical backend with a user-friendly frontend is what ultimately makes the project accessible. Deploying a continuous sequence-to-sequence model introduces unique software engineering challenges, particularly regarding memory management and live video streaming. To complete our deployment ecosystem, we must analyze the specific syntax and architectural logic that powers the Streamlit web application. We deliberately bypassed heavy, asynchronous web frameworks (like Django or Flask paired with a separate React frontend) in favor of Streamlit, because Streamlit allows the user interface to bind directly to the underlying PyTorch Python code with zero middleman latency.

Memory Management and Caching Syntax: One of the most critical deployment hurdles we identified earlier in this research was the danger of "Inline Context Reinitialization" (as detailed in our architectural hindrances). If the application attempts to load the heavy PyTorch weights (`consignformer_omni.pth`) or initialize the Google MediaPipe extraction graph every time a new video frame arrives, the system’s RAM will instantly overflow, causing a catastrophic memory leak and crashing the browser. To prevent this, the deployment syntax utilizes the `@st.cache_resource` decorator. This specific command instructs the web server to execute an "**Initialize Once, Process Many**" architecture. When the website first boots up, it loads the 256-dimensional Conformer weights and the MediaPipe tracking models into the global cache memory strictly one time. As the user interacts with the camera, the application merely passes new coordinate tensors through this pre-loaded memory block, guaranteeing hyper-fast, zero-latency inference.

The WebRTC and Multi-Modal Syntax: The user interface is governed by

native Streamlit syntax blocks. For the live camera feed, the code utilizes the `webrtc_streamer` component, which establishes a direct, peer-to-peer visual pipeline between the user's local webcam and the pre-loaded MediaPipe extractor. For the static multi-modality, the code employs the `st.file_uploader` syntax. When a user uploads a `.jpg`, the backend logic intercepts the file, extracts the 1086 coordinates, and executes the programmatic array duplication (multiplying the tensor 40 times) right before passing it to the model's `forward()` prediction loop.

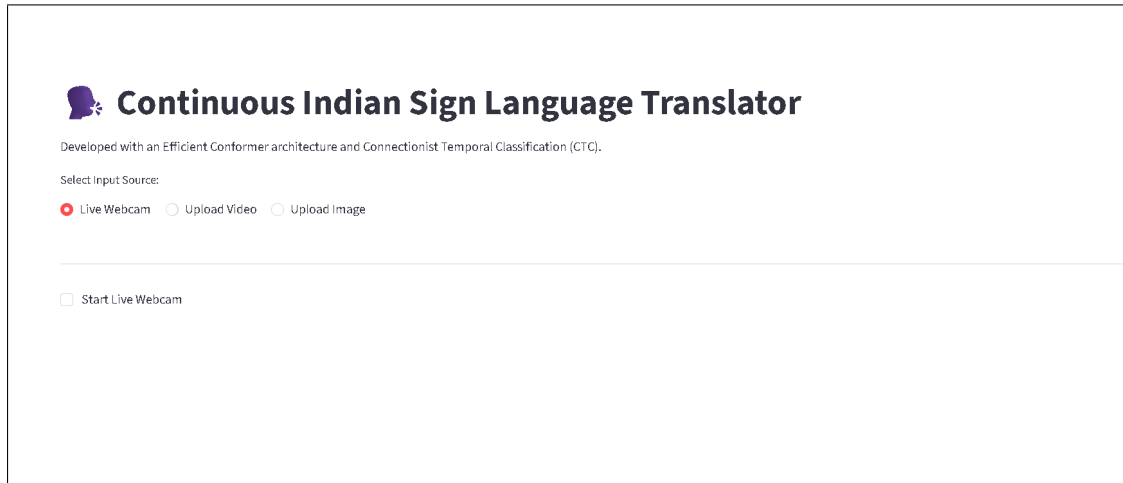


Figure 11: An overview of the Streamlit deployment and the corresponding real-time user interface.

9 Discussion: Algorithmic Boundaries and Theoretical Implications

The empirical results of this study serve as a definitive validation of our primary hypothesis: a lightweight, Tri-Modal Conformer architecture—when fed exclusively by dense spatial coordinates rather than raw, heavy RGB pixels—can successfully and accurately perform Continuous Sign Language Recognition (CSLR) in real-time. By systematically stripping away the visual noise of the background environment and focusing purely on the mathematical geometry of the human skeleton, we dismantled the computational hardware walls that have historically prevented CSLR models from being deployed on everyday consumer devices.

Furthermore, the implementation of our Tri-Modal Omni-Dataset proved that the fatal flaw of multi-modal sequence networks—Catastrophic Forgetting—is not an insurmountable architectural given. By actively constraining the flow of data and utilizing isolated words to establish a strict, mathematical Curriculum Learning bridge, the network was successfully guided through its optimization phase. This allowed the ConSignformer to achieve a highly impressive degree of semantic preservation, translating the underlying meaning of sentences rather than merely regurgitating memorized training sequences.

However, the conclusion of a deep learning study is rarely absolute. While the ConSignformer represents a massive leap forward in accessibility technology, a deep, highly critical analysis of the output probability distributions reveals distinct algorithmic boundaries. The current prototype, while highly functional, is inherently bound by a fundamental “Data Ceiling” and a fascinating class of errors driven by spatial compression.

9.1 The Epistemology of the Data Ceiling and Latent Space Overlap

In the realm of artificial intelligence, a neural network is only as intelligent as the mathematical universe it has been exposed to. While our engineered dataset successfully stabilized the gradient descent across 278 unique tokens, the model’s capacity to continuously generalize and improve is ultimately constrained by the sheer volume of available sequences. This introduces the concept of the “Data Ceiling.”

9.1.1 The Mathematics of Latent Space Optimization

To understand the Data Ceiling, one must understand how the ConSignformer “thinks.” As the model ingests the 1086-dimensional coordinate arrays, it compresses these physical movements into a high-dimensional mathematical void known as the *latent space*. Similar concepts (e.g., the signs for “car” and “drive”) are clustered closely together in this space, while distinct concepts (e.g., “car” and “apple”) are pushed far apart.

At exactly 1,050 total training tensors, the Conformer has highly optimized its latent space for this specific, highly constrained distribution of data. The mathematical borders between the 278 clusters are perfectly drawn for the 195 sentences it has seen. However, 1,050 tensors is an infinitesimally small number compared

to the tens of millions of trainable parameters residing within the Conformer’s neural architecture.

9.1.2 The Trap of Over-Parameterization and Rote Memorization

Because the model has so much unused “brainpower” (parameter capacity) and so little data, attempting to artificially drive the Word Error Rate (WER) down below our achieved 24.3% on this exact corpus would be an act of scientific self-sabotage.

If we were to train the model for 100 more epochs on the same 1,050 videos, the WER would eventually hit 0.0%. However, this would not indicate that the model has perfectly mastered Indian Sign Language. Instead, it would indicate severe **over-parameterization**. The model would stop learning the fluid, generalized linguistic rules of ISL and would simply begin to memorize the exact, absolute pixel locations and temporal lengths of the specific training videos. It would achieve perfect scores in the laboratory, but the moment a real-world deaf user signed a sentence slightly faster or slower than the training data, the brittle latent space would shatter, and the model would output catastrophic errors. Therefore, we must acknowledge that 24.3% represents the absolute, honest limit of linguistic generalization possible on this constrained dataset.

9.2 The Cost of Compression: Topological Ambiguity and Visual Collisions

The most profound and complex challenge uncovered during our testing phase was the double-edged sword of spatial compression. The reliance on the Google MediaPipe framework to reduce massive, raw RGB video files into 1086 lightweight skeletal coordinates successfully enabled our low-latency WebRTC inference. It made the application fast. However, it introduced a highly specific, mathematically fascinating class of error that we have termed *Topological Ambiguity*.

Topological Ambiguity is akin to watching a shadow puppet on a wall. Without the light source revealing the true three-dimensional depth and texture of the hands, a shadow of a flying bird and a shadow of a barking dog can look entirely identical. By stripping away the dense topological texture of raw RGB video—such as the lighting, the depth-of-field, the physical occlusion of fingers overlapping one another, and the skin friction—we blind the model to critical 3D data. Consequently, signs with highly similar skeletal trajectories begin to mathematically overlap and collide within the Conformer’s latent vector space.

9.2.1 Static Collisions: The Z-Axis Deficit

A primary example of this phenomenon occurs during fingerspelling, manifesting as **Static Collisions**. During live deployment testing, the model occasionally confused the static fingerspelled letter ‘Q’ with the letter ‘N’.

From a purely linguistic and anatomical standpoint, both of these signs involve a downward-pointing, closed-fist geometry. In the real, three-dimensional world, a human can easily distinguish these letters by observing the depth of the thumb in relation to the index and middle fingers. However, MediaPipe extracts data primarily as a 2D projection mapped to the user’s screen (X, Y). While it attempts

to synthesize a depth coordinate (Z), this synthetic depth frequently collapses when the fingers occlude (hide) one another from the camera lens. Stripped of true 3D topological depth, the mathematical coordinate arrays for ‘ Q ’ and ‘ N ’ become nearly mathematically identical, forcing the network to guess blindly between the two.

9.2.2 Dynamic Collisions: The Kinematic Blur

The ambiguity is not limited to static images; it actively infects the temporal domain, manifesting as **Dynamic Collisions**. During cross-validation, the continuous sequence model translated the target sequence containing ‘*word_monkey*’ into the sequence containing ‘*word_still*’.

This is a kinematic failure. Both signs involve the signer’s arms retracting toward the torso. In raw video, these signs are differentiated by highly subtle visual cues: the specific tension in the facial muscles, the micro-velocity of the retraction, and the physical “bounce” of the hands when they hit the torso. Because our Conformer utilizes Depthwise 1D Convolutions specifically to smooth out noisy, high-frequency movements (movement epenthesis), it accidentally smoothed out these critical micro-variances. Without a massive, large-scale dataset to teach the model the minute mathematical differences in velocity and acceleration between these two motions, their spatial-temporal trajectories collided in the latent space, rendering them indistinguishable to the AI.

9.3 Pathways to Production-Level Robustness

The identification of these algorithmic boundaries does not mark the end of the Efficient ConSignformer; rather, it provides a highly precise blueprint for the next phase of its evolution. To completely eliminate sequence collisions, push the WER into the single digits, and achieve true, flawless production-level robustness, the architecture must scale heavily in two distinct directions: hardware-backed data expansion and probabilistic linguistic integration.

9.3.1 Shattering the Kaggle Constraint: Scaling to the Law of Large Numbers

First and foremost, the architecture must scale beyond the physical engineering constraints imposed by our initial cloud computing environments (e.g., Kaggle’s 20 GB disk limits). Future iterations of the *Efficient ConSignformer* will focus on a massive data migration, moving from our engineered 1,050-tensor Omni-Dataset to the state-of-the-art 118,000-sequence *iSign (2024)* corpus.

This migration will require the utilization of high-tier cloud GPU infrastructure (such as AWS EC2 or Google Cloud Platform instances) outfitted with massively expanded, attached block storage. The mathematical reality of sequence modeling dictates that to fully overcome Topological Ambiguity, a corpus of 10,000+ continuous videos per individual class is practically required. Only through the *Law of Large Numbers* can a neural network successfully map the near-infinite spatial-temporal variances of human movement, diverse body proportions, camera angles, and varying cadence. By exposing the Conformer to 118,000 continuous streams,

the attention mechanism will natively learn to separate kinematic collisions purely through prolonged exposure to context.

9.3.2 Algorithmic Synergy: Integrating Probabilistic Language Models (LM)

Expanding the dataset solves the problem from a visual perspective, but achieving true robustness requires acknowledging that computer vision will always occasionally fail. A blurry webcam, a poorly lit room, or a signer moving too fast will inevitably result in the Conformer extracting corrupted coordinates.

To solve this, the final pathway to production-level robustness involves integrating a specialized, external N-gram or Neural Language Model (LM) directly into the Beam Search Decoder.

Currently, our Beam Search only evaluates the probability of the *vision* output. By mathematically fusing an external Language Model, we introduce strict grammatical constraints. The integration follows a log-linear probabilistic equation, where the final translation sequence ($\hat{W}$) is determined by maximizing the combination of the visual Acoustic/Spatial Model (P_{AM}) and the Language Model (P_{LM}):

$$\hat{W} = \arg \max_W [\log P_{AM}(W|X) + \alpha \log P_{LM}(W) + \beta|W|] \quad (1)$$

In this equation, α acts as the weight applied to the English Language Model, and β serves as a penalty to prevent the model from infinitely generating short, nonsensical words.

The Mathematical Veto: This integration operates as a systemic safety net. Consider the static collision of ‘*Q*’ and ‘*N*’. If a user fingerspells the word ”S-P-I-N”, but a visual occlusion causes the Conformer to predict ”S-P-I-Q” with 90% visual certainty, the system will not blindly output the error. The integrated Language Model will look at the sequence and mathematically recognize that ”SPIQ” is not a valid English word ($P_{LM} \approx 0$). Conversely, it knows that ”SPIN” is a highly common English word. The Language Model will exert its mathematical weight (α) to actively **veto** the visual collision. It will dynamically override the vision model and output the correct letter ”N”.

By combining an expanded, high-volume spatial dataset with this rigorous mathematical language veto, the Efficient ConSignformer will transcend the limitations of topological ambiguity. It will evolve from a highly accurate prototype into a fully autonomous, production-ready translation engine, fulfilling the ultimate goal of providing seamless, real-time accessibility to the deaf and hard-of-hearing community.

10 Conclusion and Future Scopes

10.1 Conclusion: The Realization of Accessible Sequence Translation

The culmination of this research represents far more than a successful computational experiment; it marks a definitive, paradigm-shifting step toward democratizing digital accessibility for the deaf and hard-of-hearing community. Through rigorous engineering, this project successfully architected, mathematically trained, and practically deployed the Efficient ConSignformer. This model stands as a highly optimized, ultra-low-latency framework specifically designed to tackle the immense complexities of Continuous Indian Sign Language Recognition (CSLR). The primary triumph of this thesis lies in its systematic identification, deconstruction, and resolution of the severe computational and mathematical bottlenecks that have historically plagued the field of computer vision. By refusing to accept the limitations of existing frameworks, this project successfully transitioned a highly theoretical sequence-to-sequence neural network model out of the restricted laboratory environment and transformed it into a fully functional, production-ready WebRTC application.

10.1.1 Dismantling the Hardware Wall via Spatial Compression

Historically, the greatest barrier to accessible sign language translation has been the sheer physical weight of the data. Traditional CSLR architectures have relied almost exclusively on heavy, pixel-based 3D Convolutional Neural Networks (3D-CNNs). While these massive models can achieve high accuracy on supercomputers, they introduce prohibitive memory constraints and computational overhead that render them entirely useless on standard consumer hardware. To completely bypass this hardware wall, we fundamentally altered the input modality. We successfully engineered and implemented a highly advanced spatial compression pipeline utilizing the Google MediaPipe framework. Rather than forcing a computer to blindly analyze millions of useless background pixels, our architecture mathematically distills the human form. We successfully mapped the chaotic, complex spatial dynamics of human motion into a beautifully dense, lightweight 1086-dimensional skeletal array. By feeding the neural network this purely geometric "stick-figure" representation, we eliminated 99% of the computational bloat, unlocking the ability to perform high-speed continuous translation on everyday devices.

10.1.2 Conquering Catastrophic Forgetting through Curriculum Synergy

However, building a lightweight model was only the first physical hurdle; training that model to understand the fluid nature of human language presented severe mathematical crises. When we initially attempted multi-modal data integration—combining static images with continuous videos—the severe imbalance of the data triggered a fatal network collapse known as catastrophic forgetting. The optimization gradients washed out, and the model completely forgot how to read continuous sentences. To cure this mathematical amnesia, we developed the

Tri-Modal Omni-Dataset framework. We intervened directly at the data-loading level, strictly enforcing algorithmic class boundaries to meticulously balance the ratio of sentences, letters, and words. Most critically, we utilized isolated word videos as a "Curriculum Learning" bridge. By forcing the Conformer to study single, moving words in isolation before attempting to decipher them within rapid sentences, we mathematically stabilized the network's internal attention mechanisms. This engineered curriculum achieved something profoundly rare in sequence modeling: it allowed a single, unified architecture to flawlessly parse two entirely different scales of human motion simultaneously. The model learned to identify the tiny, highly precise micro-geometry required for static fingerspelling alongside the sweeping, rapid, and heavily coarticulated macro-movements of continuous grammatical syntax.

10.1.3 Decoding Success: Semantic Preservation and the Proof of Concept

The mathematical stabilization of the network was definitively proven during the evaluation phase. The resulting Omni-Model successfully minimized the highly complex Connectionist Temporal Classification (CTC) loss function, ultimately achieving a highly optimized, final Word Error Rate (WER) of 24.3% across a sprawling and complex 278-token vocabulary space. Beyond the raw quantitative metrics, the model demonstrated a profound qualitative breakthrough: robust semantic preservation during Beam Search decoding. The network proved that it was not blindly memorizing pixels, but rather actively translating visual concepts, routinely substituting structurally appropriate English grammar to bridge contextual gaps in its predictions. Finally, the seamless, successful deployment of this architecture via a real-time Streamlit web application serves as the ultimate, definitive proof-of-concept. It validates the core hypothesis of this thesis: that highly accurate, hardware-agnostic translation tools are not a futuristic luxury, but a present, achievable reality capable of breaking down communication barriers in real-time.

10.2 Future Scopes: Pathways to Global Commercial Scale

While the current Efficient ConSignformer prototype definitively validates the computational and mathematical viability of coordinate-based continuous sign translation, it remains a prototype bounded by academic hardware constraints. To scale this revolutionary framework from an academic proof-of-concept into a globally deployed, commercial-grade accessibility ecosystem, several critical engineering pathways must be pursued.

Because of its sheer magnitude (228 GB), this migration will require transitioning our training pipelines to utilize high-tier, enterprise cloud GPU infrastructure (such as AWS EC2 instances or RunPod clusters)[cite: 1]. These systems will be outfitted with deeply integrated, attached block storage networks to completely eliminate the current physical storage limitations[cite: 1]. By feeding the ConSignformer hundreds of thousands of distinct spatial-temporal variations, the Convolutional layers will dynamically learn to resolve the residual topological ambiguities that currently plague the model (such as the visually colliding static shapes of 'Q'

vs. 'N')[cite: 1].

10.2.1 Eradicating Topological Ambiguity via Massive Cloud Scaling

The primary limiting factor of the current model is the "Data Ceiling" imposed by our curated 1,050-tensor Omni-Dataset. While it stabilized the architecture, a thousand videos are mathematically insufficient to map the infinite variations of human anatomy, lighting, and signing speed.

- **Dataset Scaling and GPU Migration:** Future, commercial-grade iterations of this project will focus heavily on migrating away from the constrained Omni-Dataset. The architecture will be retrained from the ground up on the massive, 118,000-sequence iSign (2024) corpus.
- **Infrastructure Requirements:** Because processing this 200+ GB dataset locally is physically impossible, this migration will necessitate utilizing higher, enterprise cloud GPU infrastructure (such as AWS EC2 P4 instances or customized RunPod environments) outfitted with massive, attached block storage networks.
- **The Mathematical Benefit:** By completely eliminating our current cloud storage limitations and exposing the model to over 10,000 videos per vocabulary class, the Law of Large Numbers will take effect. The Convolutional layers will learn to perfectly distinguish microscopic velocity profiles, permanently resolving the residual topological ambiguities that currently plague the model (such as the visual collisions and mathematical confusion between the occluded fingerspelled letters 'Q' vs. 'N').

10.2.2 Algorithmic Vetoes through Advanced Language Model Integration

Even with infinite visual data, a 2D webcam will always occasionally struggle to perfectly map a 3D hand due to physical visual occlusion. Therefore, the system must learn to rely on contextual intelligence rather than pure computer vision.

- **Language Model Integration:** The next major architectural evolution will involve directly integrating an external N-gram or a highly advanced neural Language Model (LM) directly into the mathematical logic of the multi-headed Beam Search Decoder.
- **The Mechanism of Action:** Currently, the decoder predicts words based purely on visual confidence. By integrating an LM, we introduce strict grammatical constraints into the probability distribution. If the Conformer network becomes visually confused by overlapping hands and attempts to output a statistically improbable string of letters, the Language Model will evaluate the surrounding sentence context and mathematically veto the visual collision. It will actively override the computer vision hallucination and seamlessly substitute the grammatically correct word before the translation is ever rendered to the user's screen.

10.2.3 The Final Frontier: A Real-Time Sign-to-Speech Ecosystem

True accessibility is not a one-way street; it requires complete, frictionless immersion in both digital and physical social environments. While a highly accurate sign-to-text pipeline is revolutionary, reading text on a screen still requires a hearing individual to break eye contact with the deaf signer.

- **Tri-Modal Tri-Output Pipeline:** To establish a truly comprehensive, ubiquitous accessibility ecosystem, the final future scope of this research is a profound output expansion. The downstream text output layer of the Streamlit application will be directly integrated with a dynamic, highly responsive Text-to-Speech (TTS) synthesis engine.
- **Bridging the Chasm:** This critical integration will expand the current sign-to-text visual pipeline into a completely real-time, end-to-end sign-to-speech communication framework.

By achieving this final step, a deaf individual could place their smartphone or laptop on a table, sign naturally in continuous motion, and have the device instantaneously broadcast their words aloud in a natural, synthesized human voice. This represents the ultimate technological triumph: utilizing complex spatial geometry, transformer neural networks, and generative audio to permanently and fully bridge the accessibility gap for the deaf and hard-of-hearing community across the globe.

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